

Siemens PTI Report R050-21

MISO DPP 2018 April West Area Phase 3 Study

Prepared for

MISO

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Executive Summary

This report presents results of a System Impact Study (SIS) performed to evaluate the interconnection of the DPP 2018 April Phase 3 West Area Group (DPP West Area) generating facilities. The results for 2024 scenario are summarized below.

1.1 Project List

The DPP West Area study group has twenty-two (22) generation projects with a combined nameplate rating of 2567 MW. The DPP West Area generating facilities are listed in Table ES-1. The modeling details and projects' slider diagrams are shown in Appendix B.

Table ES-1: Generating Facilities in DPP 2018 April West Area Group

MISO Project #	Service Type	TO	County	State	Point of Interconnection	Fuel Type	ERIS Output	NRIS Output	SH MW	SPK MW
J953	External NRIS	ITCM	Johnson	IA	AMIL.IOW_AFRYT	Diesel	1.83	1.83	0.0	1.83
J954	External NRIS	ITCM	Johnson	IA	AMIL.IOW_AFRYT	Solar	1.4	1.4	0.7	1.4
J963	NRIS	ITCM	Cedar	IA	Bennett - Graham 69 kV	Diesel	9	9	0.0	9.0
J967	ERIS	Xcel	Mower	MN	Adams 345 kV	Wind	150	0	150.0	23.4
J975	ERIS	OTP	Cass	ND	Buffalo 115 kV	Wind	150	0	150.0	23.4
J982	ERIS	MEC	Dickinson, Emmet	IA	Obrien County - Kossuth 345 kV	Wind	300	0	300.0	46.8
J1040	NRIS	MDU	McIntosh	ND	Wishek Junction 230 kV	Wind	250	250	250.0	39.0
J1045	NRIS	Xcel	Murray	MN	Fenton - Chanarambie 115 kV	Battery	20	20	20.0	20.0
J1050	NRIS	ITCM	Boone, Hamilton	IA	Doud Tap 161 kV	Wind	225	225	225.0	35.1
J1072	NRIS	Xcel	Mower	MN	Adams 345 kV	Solar	150	150	75.0	150.0
J1084	NRIS	ITCM	Clinton	IA	Rock Creek 345 kV	Solar	150	150	75.0	150.0
J1092	NRIS	Xcel	Saint Croix	WI	Three Lakes 115 kV	Solar	100	100	50.0	100.0
J1110	NRIS	SMMPA	Mower	MN	North Austin 161 kV	Solar	100	100	50.0	100.0
J1124	NRIS	SMMPA	Olmsted	MN	Byron 345 kV	Solar	100	100	50.0	100.0
J1131	NRIS	MEC	Scott	IA	Sub 56 161 kV	Solar	100	100	50.0	100.0
J1132	ERIS	ITCM	Union	IA	Creston East 69 kV	Solar	50	0	25.0	50.0
J1135	NRIS	ITCM	Des Moines	IA	Huntwoods 69 kV	Solar	50	50	25.0	50.0

MISO Project #	Service Type	TO	County	State	Point of Interconnection	Fuel Type	ERIS Output	NRIS Output	SH MW	SPK MW
J1164	ERIS	ITCM	Rock	MN	Magnolia 161 kV	Solar	80	0	40.0	80.0
J1169	ERIS	Xcel	McCook	SD	Grant 115 kV	Solar	50	0	25.0	50.0
J1174	NRIS	ITCM	Worth	IA	Bison - Colby 345 kV	Solar	165	165	82.5	165.0
J1175	ERIS	ITCM	Worth	IA	Bison - Colby 345 kV	Wind	165	0	82.5	165.0
J1181	ERIS	ITCM	Chickasaw	IA	Hazleton - Mitchell county 345 kV	Wind	200	0	200.0	31.2

1.2 Reactive Power Requirements for Non-Synchronous Generation (FERC Order 827)

For non-synchronous generation projects in the DPP 2018 April West Area study group, if they do not have signed Generator Interconnection Agreement (GIA) or Provisional GIA (PGIA) by September 21, 2016, they are required to provide dynamic reactive power within the range of 0.95 leading to 0.95 lagging at the high-side of the generator substation.

All non-synchronous generation projects in this study group are required to meet the reactive power requirements per FERC Order 827.

The reactive power requirement analysis results are summarized as following:

- J982, J1072, J1084, J1110, J1124, and J1164 do not meet the reactive power requirements per FERC Order 827.
- All other non-synchronous generation projects can meet the reactive power requirements per FERC Order 827.

1.3 Total Network Upgrades for all Projects

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Network Resource Interconnection Service as of the System Impact Study report date. The total cost of network upgrades in the interconnection plan required for each generation project is listed in Table ES-2. The costs for Network Upgrades are planning level estimates and subject to be revised in the facility studies.

Table ES-2: Total Cost of Network Upgrades for DPP 2018 April West Area Generation Projects

Project Num	ERIS Network Upgrades (\$)											NRIS Network Upgrades (\$)	Inter-connection Substation TO NUs (\$)	TO's Inter-connection Facilities (TOIF)	SNU (\$)	Total Network Upgrade Cost (Exclude TOIF & Affected System) (\$)
	MWEX Voltage Stability	MISO Thermal & Voltage	Transient Stability	Short Circuit	MDU LPC	OTP LPC	CIPCO AFS	MPC AFS	PJM AFS	AECI AFS	SPP AFS					
J953	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
J954	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
J963	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
J967	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$4,631,343	\$0	\$928,500	\$1,705,500	\$0	\$928,500
J975	\$0	\$3,386,441	\$0	\$0	\$0	\$0	\$0	\$20,761,954	\$0	\$0	\$6,336,633	\$0	\$479,091	\$1,043,191	\$0	\$3,865,532
J982	\$0	\$3,184,746	\$0	\$0	\$0	\$0	\$0	\$0	\$8,130,000	\$0	\$17,716,774	\$0	\$14,000,000	\$1,250,000	\$0	\$17,184,746
J1040	\$0	\$11,551,525	\$0	\$0	\$300,000	\$0	\$0	\$33,038,046	\$0	\$0	\$11,102,024	\$0	\$397,037	\$397,037	\$0	\$12,248,562
J1045	\$0	\$144,068	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$1,426,716	\$0	\$0	\$0	\$0	\$144,068
J1050	\$0	\$364,407	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$408,508	\$403,313	\$0	\$772,915
J1072	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$4,631,343	\$0	\$928,500	\$1,705,500	\$0	\$928,500
J1084	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$12,390,000	\$0	\$0	\$0	\$1,294,462	\$661,704	\$0	\$1,294,462
J1092	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$3,476,653	\$0	\$1,511,000	\$1,125,000	\$0	\$1,511,000
J1110	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$3,383,313	\$6,000,000	\$4,761,550	\$341,450	\$0	\$10,761,550
J1124	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$3,401,667	\$0	\$1,594,400	\$1,097,700	\$0	\$1,594,400
J1131	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$825,000	\$1,200,000	\$0	\$825,000
J1132	\$0	\$50,847	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$227,244	\$487,844	\$0	\$278,091
J1135	\$0	\$8,475	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$558,384	\$0	\$8,475
J1164	\$0	\$211,864	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$6,320,652	\$0	\$6,908,747	\$396,282	\$0	\$7,120,611
J1169	\$0	\$144,068	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$4,420,468	\$0	\$5,485,000	\$1,054,000	\$0	\$5,629,068
J1174	\$0	\$84,746	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$6,897,617	\$0	\$5,797,466	\$461,139	\$0	\$5,882,212
J1175	\$0	\$228,814	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$6,897,617	\$0	\$5,797,466	\$461,139	\$0	\$6,026,280
J1181	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$7,510,000	\$0	\$4,668,050	\$0	\$12,653,333	\$892,278	\$0	\$12,653,333
Total (\$)	\$0	\$19,360,000	\$0	\$0	\$300,000	\$0	\$0	\$53,800,000	\$28,030,000	\$0	\$85,310,870	\$6,000,000	\$63,997,304	\$15,241,461	\$0	\$89,657,304

The study was performed under the direction of MISO by Siemens PTI and an ad hoc study group. The ad hoc study group was formed to review the study scope, methodology, models, and results. The ad hoc study group consisted of representatives from the interconnection customers and the following utility companies – Ameren, American Transmission Company, Basin Electric Power, Cedar Falls Utilities, Central Iowa Power Cooperative, City of Springfield (IL) Water Light & Power, Columbia (MO) Water and Light, Commonwealth Edison, Corn Belt Power Cooperative, Dairyland Power, Great River Energy, ITC Midwest, Lincoln Electric System, Manitoba Hydro, MidAmerican Energy Company, MISO, Minnesota Power, Minnkota Power, Missouri River Energy Services, Montana-Dakota Utilities Co., Muscatine Power & Water, Nebraska Public Power District, Northwestern Public Service, Omaha Public Power District, Otter Tail Power, PJM, Southern Illinois Power Cooperative, Southern Minnesota Municipal Power Agency, SPP, Western Area Power Administration, and Xcel Energy.

1.4 Per Project Summary

This section provides estimated cost of Network Upgrades on a per project basis for the 2024 scenario. The shared cost of Network Upgrades for all the generation projects are listed below.

The Interconnection Customers are required to mitigate the constraints observed from the 2024 scenario.

1.4.1 J953 Summary

Network Upgrade	Cost	J953	NUs Type	Self Funding?
No Network Upgrades		\$0		
Total Cost Per Project for Actual NRIS Elections for each Project		\$0		

1.4.2 J954 Summary

Network Upgrade	Cost	J954	NUs Type	Self Funding?
No Network Upgrades		\$0		
Total Cost Per Project for Actual NRIS Elections for each Project		\$0		

1.4.3 J963 Summary

Network Upgrade	Cost	J963	NUs Type	Self Funding?
No Network Upgrades		\$0		
Total Cost Per Project for Actual NRIS Elections for each Project		\$0		

1.4.4 J967 Summary

Network Upgrade	Cost	J967	NUs Type	Self Funding?
SPP ERIS Network Upgrades	\$85,927,678	\$4,631,343	SPP	No
Total Cost Per Project for Actual ERIS Elections for each Project		\$4,631,343		

1.4.5 J975 Summary

Network Upgrade	Cost	J975	NUs Type	Self Funding?
CSLTNET-Mapleton 115 kV	\$200,000	\$200,000	MISO SH	Yes
Buffalo 345-230-41.6 kV xfmr #2	\$3,000,000	\$3,000,000	MISO SH	Yes
2x75 Mvar switched capacitor at Cedar Mountain 345 kV (615647)	\$3,000,000	\$186,441	MISO Voltage	Yes
Wilton-Winger 230 kV line	\$500,000	\$206,016	MPC	No
1x75 MVar MSC at Bison 345	\$1,500,000	\$390,582	MPC	No
2x40 MVar MSC at Maple River 230	\$2,000,000	\$958,199	MPC	No
+200 MVar SVC at Winger 230	\$30,000,000	\$12,259,887	MPC	No
+50 MVar SVC at Frontier 230	\$15,000,000	\$6,129,944	MPC	No
2x30 MVar MSC at Frontier 230	\$2,000,000	\$817,326	MPC	No
SPP ERIS Network Upgrades	\$85,927,678	\$6,336,633	SPP	No
Total Cost Per Project for Actual ERIS Elections for each Project		\$30,485,028		

1.4.6 J982 Summary

Network Upgrade	Cost	J982	NUs Type	Self Funding?
J748 POI-Raun 345 kV	\$1,500,000	\$1,500,000	MISO SH	Yes
Rutland-Fox Lake 161 kV	\$1,100,000	\$1,100,000	MISO SH	ITCM: Yes SMMPA: No
2x75 Mvar switched capacitor at Cedar Mountain 345 kV (615647)	\$3,000,000	\$584,746	MISO Voltage	Yes
Nelson;B- Electric JCT;R 345 kV	\$36,200,000	\$8,130,000	PJM	No
SPP ERIS Network Upgrades	\$85,927,678	\$17,716,774	SPP	No
Total Cost Per Project for Actual ERIS Elections for each Project		\$29,031,520		

1.4.7 J1040 Summary

Network Upgrade	Cost	J1040	NUs Type	Self Funding?
Ellendale-Aberdeen 115 kV	\$60,000	\$60,000	MISO SH, MDU LPC	MDU: Yes
Add a breaker at Merricourt which will eliminate tripping of both 230 kV lines for the "P23:230:MDU::MERRICOURT::2722" contingency	\$1,500,000	\$1,500,000	MISO SH, MDU LPC	Yes
Merricourt-Wishek 230 kV	\$9,000,000	\$9,000,000	MISO SH	Yes
2x75 Mvar switched capacitor at Cedar Mountain 345 kV (615647)	\$3,000,000	\$991,525	MISO Voltage	Yes
Ellendale-Aberdeen 115 kV (MDU LPC)	\$250,000	\$250,000	MDU LPC	MDU: Yes
J302 POI-Mandan 230 kV (MDU LPC)	\$50,000	\$50,000	MDU LPC	Yes
Wilton-Winger 230 kV line	\$500,000	\$293,984	MPC	No
Drayton 230-115 kV xfmr	\$2,100,000	\$2,100,000	MPC	No
Prairie-Walle 230 kV line	\$200,000	\$200,000	MPC	No
Winger-Walle 230 kV line	\$500,000	\$500,000	MPC	No
1x75 MVAR MSC at Bison 345	\$1,500,000	\$1,109,418	MPC	No
2x40 MVAR MSC at Maple River 230	\$2,000,000	\$1,041,801	MPC	No
+200 MVAR SVC at Winger 230	\$30,000,000	\$17,740,113	MPC	No
+50 MVAR SVC at Frontier 230	\$15,000,000	\$8,870,056	MPC	No
2x30 MVAR MSC at Frontier 230	\$2,000,000	\$1,182,674	MPC	No
SPP ERIS Network Upgrades	\$85,927,678	\$11,012,024	SPP	No
SPP NRIS Network Upgrades	\$90,000	\$90,000	SPP	No
Total Cost Per Project for Actual NRIS Elections for each Project		\$55,991,595		

1.4.8 J1045 Summary

Network Upgrade	Cost	J1045	NUs Type	Self Funding?
2x75 Mvar switched capacitor at Cedar Mountain 345 kV (615647)	\$3,000,000	\$144,068	MISO Voltage	Yes
SPP ERIS Network Upgrades	\$85,927,678	\$1,426,716	SPP	No
Total Cost Per Project for Actual NRIS Elections for each Project		\$1,570,784		

1.4.9 J1050 Summary

Network Upgrade	Cost	J1050	NUs Type	Self Funding?
2×75 Mvar switched capacitor at Cedar Mountain 345 kV (615647)	\$3,000,000	\$364,407	MISO Voltage	Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$364,407		

1.4.10 J1072 Summary

Network Upgrade	Cost	J1072	NUs Type	Self Funding?
SPP ERIS Network Upgrades	\$85,927,678	\$4,631,343	SPP	No
Total Cost Per Project for Actual NRIS Elections for each Project		\$4,631,343		

1.4.11 J1084 Summary

Network Upgrade	Cost	J1084	NUs Type	Self Funding?
Nelson;B- Electric JCT;R 345 kV	\$36,200,000	\$12,390,000	PJM	No
Total Cost Per Project for Actual NRIS Elections for each Project		\$12,390,000		

1.4.12 J1092 Summary

Network Upgrade	Cost	J1092	NUs Type	Self Funding?
SPP ERIS Network Upgrades	\$85,927,678	\$3,476,653	SPP	No
Total Cost Per Project for Actual NRIS Elections for each Project		\$3,476,653		

1.4.13 J1110 Summary

Network Upgrade	Cost	J1110	NUs Type	Self Funding?
Upgrade Pleasant Valley 345/161 kV Transformer #1	\$3,000,000	\$3,000,000	NRIS	Yes
Upgrade Pleasant Valley 345/161 kV Transformer #2	\$3,000,000	\$3,000,000	NRIS	Yes
SPP ERIS Network Upgrades	\$85,927,678	\$3,383,313	SPP	No
Total Cost Per Project for Actual NRIS Elections for each Project		\$9,383,313		

1.4.14 J1124 Summary

Network Upgrade	Cost	J1124	NUs Type	Self Funding?
SPP ERIS Network Upgrades	\$85,927,678	\$3,401,667	SPP	No
Total Cost Per Project for Actual NRIS Elections for each Project		\$3,401,667		

1.4.15 J1131 Summary

Network Upgrade	Cost	J1131	NUs Type	Self Funding?
No Network Upgrades		\$0		
Total Cost Per Project for Actual NRIS Elections for each Project		\$0		

1.4.16 J1132 Summary

Network Upgrade	Cost	J1132	NUs Type	Self Funding?
2×75 Mvar switched capacitor at Cedar Mountain 345 kV (615647)	\$3,000,000	\$50,847	MISO Voltage	Yes
Total Cost Per Project for Actual ERIS Elections for each Project		\$50,847		

1.4.17 J1135 Summary

Network Upgrade	Cost	J1135	NUs Type	Self Funding?
2×75 Mvar switched capacitor at Cedar Mountain 345 kV (615647)	\$3,000,000	\$8,475	MISO Voltage	Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$8,475		

1.4.18 J1164 Summary

Network Upgrade	Cost	J1164	NUs Type	Self Funding?
2×75 Mvar switched capacitor at Cedar Mountain 345 kV (615647)	\$3,000,000	\$211,864	MISO Voltage	Yes
SPP ERIS Network Upgrades	\$85,927,678	\$6,320,652	SPP	No
Total Cost Per Project for Actual ERIS Elections for each Project		\$6,532,516		

1.4.19 J1169 Summary

Network Upgrade	Cost	J1169	NUs Type	Self Funding?
2×75 Mvar switched capacitor at Cedar Mountain 345 kV (615647)	\$3,000,000	\$144,068	MISO Voltage	Yes
SPP ERIS Network Upgrades	\$85,927,678	\$4,420,468	SPP	No
Total Cost Per Project for Actual ERIS Elections for each Project		\$4,564,536		

1.4.20 J1174 Summary

Network Upgrade	Cost	J1174	NUs Type	Self Funding?
2×75 Mvar switched capacitor at Cedar Mountain 345 kV (615647)	\$3,000,000	\$84,746	MISO Voltage	Yes
SPP ERIS Network Upgrades	\$85,927,678	\$6,897,617	SPP	No
Total Cost Per Project for Actual NRIS Elections for each Project		\$6,982,363		

1.4.21 J1175 Summary

Network Upgrade	Cost	J1175	NUs Type	Self Funding?
2×75 Mvar switched capacitor at Cedar Mountain 345 kV (615647)	\$3,000,000	\$228,814	MISO Voltage	Yes
SPP ERIS Network Upgrades	\$85,927,678	\$6,897,617	SPP	No
Total Cost Per Project for Actual ERIS Elections for each Project		\$7,126,431		

1.4.22 J1181 Summary

Network Upgrade	Cost	J1181	NUs Type	Self Funding?
Nelson;B- Electric JCT;R 345 kV	\$36,200,000	\$7,510,000	PJM	No
SPP ERIS Network Upgrades	\$85,927,678	\$4,668,050	SPP	No
Total Cost Per Project for Actual ERIS Elections for each Project		\$12,178,050		

1.5 Study Compliance with NERC FAC-002-2 Standard

This DPP 2018 April West Area study was completed in compliance with NERC FAC-002-2:

R1.1: The reliability impact of the new interconnection, or materially modified existing interconnection, on affected system(s).

Section 3 covers summer peak steady-state analysis results which include thermal and voltage constraints impacted by the DPP West Area generating facilities. Thermal and voltage upgrades required to interconnect the new generating facilities are also identified.

Section 4 covers summer shoulder steady-state analysis results which include thermal and voltage constraints impacted by the DPP West Area generating facilities. Thermal and voltage upgrades required to interconnect the new generating facilities are also identified.

Section 5.1 covers reliability impact of the generating facilities per OTP Local planning Criteria (LPC). Network Upgrades required to interconnect the new generating facilities are also identified.

Section 5.2 covers reliability impact of the generating facilities per MDU Local planning Criteria (LPC). Network Upgrades required to interconnect the new generating facilities are also identified.

Section 6.1 covers reliability impact of the new generating facilities in the CIPCO affected systems.

Section 6.2 covers reliability impact of the new generating facilities in the MPC affected systems.

Section 6.3 covers reliability impact of the new generating facilities in the PJM affected systems.

Section 6.4 covers reliability impact of the new generating facilities in the AECI affected systems.

Section 6.5 covers reliability impact of the new generating facilities in the SPP affected systems.

Section 7 covers transient stability analysis results.

Section 8 covers voltage stability (PV) analysis on the MWEX System Operating Limit (SOL). Network Upgrades required for MWEX voltage stability are identified.

Section 9 covers short circuit reliability impact of the new generating facilities.

Section 10 covers Deliverability reliability impact of the new NRIS generating facilities.

R1.2: Adherence to applicable NERC Reliability Standards; regional and Transmission Owner planning criteria; and Facility interconnection requirements.

Sections 2.2-2.4, Section 5, Section 6, and Section 7 all cover NERC Reliability Standard TPL-001-4.

Section 5.1 covers OTP LPC.

Section 5.2 covers MDU LPC.

Section 6.1 covers CIPCO system planning criteria.

Section 6.2 covers MPC system planning criteria.

Section 6.3 covers PJM system planning criteria.

Section 6.4 covers AECI system planning criteria.

Section 6.5 covers SPP system planning criteria.

Section 8 (voltage stability analysis) covers individual system planning criteria (ATC).

Section 10 covers MISO system planning criteria.

R1.3: Steady-state, short-circuit, and dynamics studies, as necessary, to evaluate system performance under both normal and contingency conditions.

Section 3 and Section 4 cover MISO steady-state assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 5.1 covers OTP's LPC assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 5.2 covers MDU's LPC assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 6.1 covers CIPCO steady-state assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 6.2 covers MPC steady-state and transient stability assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 6.3 covers PJM steady-state assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 6.4 covers AECI steady-state assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 6.5 covers SPP steady-state assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 7 covers transient stability studies under NERC category P0 to P7 contingencies (TPL-001-4).

Section 8 covers steady-state voltage stability assessment.

Section 9 covers short circuit assessment.

Section 10 covers MISO deliverability study (steady-state assessment) including NERC category P0 to P1 contingencies (TPL-001-4).

R1.4: Study assumptions, system performance, alternatives considered, and coordinated recommendations. While these studies may be performed independently, the results shall be evaluated and coordinated by the entities involved.

Section 2.1, Section 2.2, Section 2.3, and Section 2.4, Section 7.2, Section 7.3, and Section 7.4 cover study assumptions and system performance criteria.

Jointly coordinated recommendations can be found in Section 5.1 (MISO and OTP), Section 5.2 (MISO and MDU), Section 6.1 (MISO and CIPCO), Section 6.2 (MISO and MPC), Section 6.3 (MISO and PJM), Section 6.4 (MISO and AECI), Section 6.5 (MISO and SPP), and Section 8 (MISO and ATC). Results in Sections 3, 4, 5, 6, 7, 9 and 10 have also been reviewed by PJM, SPP, AECI, CIPCO, MPC, and ATC.

Introduction

Twenty-two (22) generation projects, listed in Table A-1 (Appendix A.1), have requested to interconnect to the MISO transmission network in the West Area and have advanced to the Definitive Planning Phase (DPP) 2018 April Phase 3 study (DPP West Area). J967, J975, J982, J1132, J1164, J1169, J1175, and J1181 have requested Energy Resource Interconnection Service (ERIS); J953 and J954 have requested external Network Resource Interconnection Service (external NRIS); All other generating facilities have requested both ERIS and NRIS.

This report presents the study results of a System Impact Study (SIS) performed to evaluate the interconnection of the generating facilities in the DPP West Area Phase 3 study.

The study was performed under the direction of MISO by Siemens PTI and an ad hoc study group. The ad hoc study group was formed to review the study scope, methodology, models, and results. The ad hoc study group consisted of representatives from the interconnection customers and the following utility companies – Ameren, American Transmission Company, Basin Electric Power, Cedar Falls Utilities, Central Iowa Power Cooperative, City of Springfield (IL) Water Light & Power, Columbia (MO) Water and Light, Commonwealth Edison, Corn Belt Power Cooperative, Dairyland Power, Great River Energy, ITC Midwest, Lincoln Electric System, Manitoba Hydro, MidAmerican Energy Company, MISO, Minnesota Power, Minnkota Power, Missouri River Energy Services, Montana-Dakota Utilities Co., Muscatine Power & Water, Nebraska Public Power District, Northwestern Public Service, Omaha Public Power District, Otter Tail Power, PJM, Southern Illinois Power Cooperative, Southern Minnesota Municipal Power Agency, SPP, Western Area Power Administration, and Xcel Energy.

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Model Development and Study Criteria

2.1 Model Development

2.1.1 Benchmark Cases

DPP 2018 April West area power flow benchmark cases representing 2024 summer shoulder and summer peak conditions were developed from the MTEP19 models with LBA dispatch.

The benchmark cases for DPP 2018 April study were created as follows:

- MISO prior queued generation projects were modeled, and their associated Network Upgrades (NU) prior to DPP 2017 August cycle were also modeled. Network Upgrades required in DPP 2017 August West Area Phase 3 were also modeled.
- DPP 2018 April generation projects in the West Area (DPP West Area, Table A-1), Central Area (Table A-3), Michigan Area (Table A-4), and ATC Area (Table A-5) were modeled with offline status.
- For MISO generation projects, their output was sunk to the MISO North (Appendix A.3, Table A-7), where generation was scaled uniformly;
- PJM generation projects were modeled and dispatched. The generation output was sunk to the PJM market (Appendix A.4, Table A-8), where generation was scaled uniformly.
- SPP generation projects were modeled and dispatched. The generation output was sunk to the SPP market (Appendix A.5, Table A-9), where generation was scaled uniformly.
- Models were further reviewed by the Ad Hoc study members (transmission owners and customers). Model corrections and changes were made based on the comments and feedback. These modeling changes are listed in Appendix A.2.
- Adjusted Square Butte DC to match the total output of the Bison (Bison 1 to 5) and Oliver County (Oliver County 1 and 2) wind farms.
- Adjusted CU DC to match the total output of Coal Creek generation units #1 and #2.
- MHEX interface transfer level is at 1530 MW in summer shoulder and 1800 MW in summer peak cases.

2.1.2 Study Cases

Summer peak study case was created by dispatching the DPP 2018 April generation projects at the specified summer peak level from the benchmark case.

Summer shoulder study case was created by dispatching the DPP 2018 April generation projects at the specified summer shoulder level from the benchmark case.

The MISO North was used for power balance, where generation was scaled uniformly.

Due to low voltages (<0.95 p.u.) at Mingo and Red Willow 345 kV buses under system intact condition in both study and benchmark summer shoulder cases, one (1) fictitious large size SVC in SPP area (Table 2-1) were added to the summer shoulder study and benchmark cases to achieve converged power flow thermal solutions.

Table 2-1: Fictitious SVCs Added Only in Summer Shoulder Case

Location	Bus #	SVC Mvar
Mingo 345 kV	531451	200

Both study and benchmark power flow cases were solved with transformer tap adjustment enabled, area interchange disabled, phase shifter adjustment enabled and switched shunt adjustment enabled.

The interface transfer levels in the 2024 study cases are summarized in Table 2-2.

Table 2-2: Interface Transfer Levels in 2024 Study Cases

Interface	2024 SH Case (MW)	2024 SPK Case (MW)
MHEX	1529	1799
MWEX	997.6	281.4
Arrowhead PST	404.7	-36.8
J732 POI – Stone Lake 345 kV	654.3	493.6

2.2 Contingency Criteria

A variety of contingencies were considered for steady-state analysis:

- NERC Category P0 with system intact (no contingencies)
- NERC Category P1 contingencies
 - NERC Category P1 contingencies, at buses with a nominal voltage of 69 kV and above, in the following areas: CWLD (area 333), AMMO (area 356), AMIL (area 357), CWLP (area 360), SIPC (area 361), WEC (area 295), WEC MI (area 296), XCEL (area 600), MP (area 608), SMMPA (area 613), GRE (area 615), OTP (area 620), ITCM (area 627), MPW (area 633), MEC (area 635), MDU (area 661), BEPC-MISO (area 663), MHEB (area 667), DPC (area 680), ALTE (area 694), WPS (area 696), MGE (area 697), UPPC (area 698), CE(area 222), NPPD (area 640), OPPD (area 645), LES (area 650), WAPA (area 652), BEPC-SPP (area 659), AECI (area 330), MIPU(area 540), KCPL (area 541), KACY (area 542), INDN (area 545).
 - Multiple-element NERC Category P1 contingencies, in Dakotas, Illinois, Iowa, Minnesota, Missouri, and Wisconsin. These specified Category P1 contingencies are listed in Appendix A.6.
- NERC Category P2-P7 contingencies
 - Selected NERC Category P2-P7 contingencies provided by the Ad Hoc Study Group, in the study region of Dakotas, Illinois, Iowa, Minnesota, Missouri, and

Wisconsin. These specified Category P2-P7 contingencies are listed in Appendix A.6.

For all contingency and post-disturbance analyses, cases were solved with transformer tap adjustment enabled, area interchange adjustment disabled, phase shifter adjustment disabled (fixed) and switched shunt adjustment enabled.

2.3 Monitored Elements

The study area is defined in Table 2-3. Facilities in the study area were monitored for system intact and contingency conditions. Under NERC category P0 conditions (system intact) branches were monitored for loading above the normal (PSS[®]E rate A) rating. Under NERC category P1-P7 conditions, branches were monitored for loading as shown in the column labeled "Post-Disturbance Thermal Limits".

Table 2-3: Monitored Elements

Owner / Area	Monitored Facilities	Thermal Limits ¹		Voltage Limits ²	
		Pre-Disturbance	Post-Disturbance	Pre-Disturbance	Post-Disturbance
AECI	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
AMIL	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.075/0.90
AMMO	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.075/0.90
ATCLLC	69 kV and above	95% of Rate A	95% of Rate B	1.05/0.95	1.10/0.90
BEPC-MISO	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
BEPC-SPP	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
CBPC	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
CMPMA	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.07/0.90
CWLD	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
CWLP	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.075/0.90
CE	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
DPC	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
GMO	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
GRE	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.92/0.90
INDN	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
ITCM	69 kV and above	100% of Rate A	100% of Rate B	1.07/1.05/0.95	1.10/0.93
KACY	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
KCPL	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
LES	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
MDU	57 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
MEC	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.96/0.95	1.05/0.96/0.95/0.94/0.93 ³

Owner / Area	Monitored Facilities	Thermal Limits ¹		Voltage Limits ²	
		Pre-Disturbance	Post-Disturbance	Pre-Disturbance	Post-Disturbance
MHEB	69 kV and above	100% of Rate A	100% of Rate B	1.12/1.1/1.07/1.05/1.04/0.99/0.97/0.96/0.95	1.15/1.10/0.94/0.90
MP	69 kV and above	100% of Rate A	100% of Rate B	1.05/1.00	1.10/0.95
MPC	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.97	1.10/0.92
MPW	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.06/0.92
MRES	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.97	1.05/0.92
NPPD	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
OPPD	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
OTP	40 kV and above	100% of Rate A	100% of Rate B	1.07/1.05/0.97	1.10/0.92
PPI	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.075/0.90
RPU	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.92
SIPC	69 kV and above	100% of Rate A	100% of Rate B	1.07/0.95	1.09/0.91
SMPMA	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
WAPA	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
WPPI	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.1/0.9
XEL	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.05/0.92

Notes

- 1: PSS®E Rate A, Rate B or Rate C
- 2: Limits dependent on nominal bus voltage
- 3: For facilities in Cedar Falls Utilities or Ames Municipal Utilities, post-contingency voltage limits are 1.05/0.94 for >200 kV, and 1.05/0.93 for others.

2.4 Performance Criteria

A branch is considered as a thermal injection constraint if the branch is loaded above its applicable normal or emergency rating for the post-change case, and any of the following conditions are met:

- 1) the generator (NR/ER) has a larger than 20% DF on the overloaded facility under post contingent condition or 5% DF under system intact condition, or
- 2) the megawatt impact due to the generator is greater than or equal to 20% of the applicable rating (normal or emergency) of the overloaded facility, or
- 3) the overloaded facility or the overload-causing contingency is at generator's outlet, or
- 4) for any other constrained facility, where none of the study generators meet one of the above criteria in 1), 2), or 3), however, the cumulative megawatt impact of the group of study generators (NR/ER) is greater than 20% of the applicable rating, then only those study generators whose individual MW impact is greater than 5% of the applicable rating and has DF greater than 5% (OTDF or PTDF) will be responsible for mitigating the cumulative MW impact constraint.

A bus is considered a voltage constraint if both of the following conditions are met. All voltage constraints must be resolved before a project can receive interconnection service.

- 1) the bus voltage is outside of applicable normal or emergency limits for the post-change case, and
- 2) the change in bus voltage is greater than 0.01 per unit.

All DPP 2018 April West Area study generators must mitigate thermal injection constraints and voltage constraints to obtain unconditional Interconnection Service.

Further, all generators requesting Network Resource Interconnection Service (NRIS) must mitigate constraints found by using the deliverability algorithm, to meet the system performance criteria for NERC category P0-P1 events, if the constraint demonstrates an incremental flow caused by the generator equal to or greater than 5% of the generator's maximum dispatch level in each case.

2.5 Reactive Power Requirements for Non-Synchronous Generation (FERC Order 827)

For non-synchronous generation projects in the DPP 2018 April West Area study group, if they do not have signed Generator Interconnection Agreement (GIA) or Provisional GIA (PGIA) by September 21, 2016, they are required to provide dynamic reactive power within the range of 0.95 leading to 0.95 lagging at the high-side of the generator substation.

All non-synchronous generation projects in this study group are required to meet the reactive power requirements per FERC Order 827.

Collector system and shunt compensation of DPP West projects are modeled, which are listed in Appendix A.1, Table A-2. An analysis was performed to study the reactive power requirements (FERC Order 827) for the non-synchronous generation projects in the DPP 2018 April West study group. The analysis was performed as follows:

Step 1: Verify whether total dynamic reactive power (reactive power from generators and dynamic compensation devices) in the plant can meet the dynamic reactive power range of 0.95 leading to 0.95 lagging at the generator terminal bus. The verification in Step 1 was performed when generator data was submitted and modeled.

Step 2: Verify whether total reactive power (reactive power from generators, dynamic compensation devices, and static compensation devices) in the plant can meet the dynamic reactive power range of 0.95 leading to 0.95 lagging at the high-side of the generator substation. The testing procedure in Step 2 is described in the following:

- Lock the high-side of the generator substation at 1.0 pu voltage by adding a fictitious SVC. This is to ensure that the test result is not affected by the system condition.
- Lock the reactive power output of the generator to the maximum limit (Q_{max}). Make sure all shunt compensation devices within the substation are at the maximum capacitive output. Adjust transformer tap to ensure bus voltages within the substation are within 0.95 – 1.05 pu range. Measure real power and reactive power from the generator plant to the high-side of the generator substation. Calculate the power factor to verify if it meets the 0.95 lagging requirement.

- Lock the reactive power output of the generator to the minimum limit (Q_{min}). Make sure all shunt compensation devices within the substation are at the maximum inductive output. Adjust transformer tap to ensure bus voltages within the substation are within 0.95 – 1.05 pu range. Measure real power and reactive power from the generator plant to the high-side of the generator substation. Calculate the power factor to verify if it meets the 0.95 leading requirement.

Appendix C lists reactive power requirement analysis results for the DPP West generation projects. The results are summarized as following:

- J982, J1072, J1084, J1110, J1124, and J1164 do not meet the reactive power requirements per FERC Order 827.
- All other non-synchronous generation projects can meet the reactive power requirements per FERC Order 827.

Summer Peak Steady-State Analysis

Summer peak steady-state analysis was performed in summer peak scenario to identify thermal and voltage upgrades required for interconnecting the generating facilities in the DPP 2018 April West Area group to the transmission system.

3.1 Study Procedure

3.1.1 Computer Programs

Steady-state analyses were performed using PSS®E version 33.12.1 and PSS®MUST version 12.4.0.

3.1.2 Study Methodology

Summer peak power flow cases were created in the procedure as described in Section 2.1. The summer peak study case can converge under post-contingency conditions. Therefore, no fictitious SVCs or Base Case Network Upgrades (BCNUs) were modeled in the summer peak cases. Nonlinear (AC) contingency analysis was performed on the benchmark and study cases, and the incremental impact of the DPP West Area generating facilities was evaluated by comparing the steady-state performance of the transmission system in the benchmark and study cases. Network upgrades were identified to mitigate any summer peak constraints.

3.2 Contingency Analysis Results for Summer Peak Condition

The incremental impact of the proposed interconnection on individual facilities was evaluated by comparing flows and voltages between benchmark case (without DPP 2018 April projects) and study case (with DPP 2018 April projects).

3.2.1 System Intact Conditions

For NERC category P0 (system intact) conditions, no thermal constraints (Table D-1) or voltage constraints (Table D-2) were identified.

3.2.2 Post Contingency Conditions

The results in this Section are for analysis of conditions following NERC Category P1-P7 contingencies. All category P1 contingencies were converged.

For P1 contingencies, no thermal constraints (Table D-3) or voltage constraints (Table D-4) were identified.

Two category P2-P7 contingencies (Table D-7) were not converged, and their dc thermal results are listed in Table D-8. The contingency was not converged in the benchmark or study cases. No mitigation plan is required for the study projects for these contingencies.

For P2-P7 contingencies, no thermal constraints (Table D-5) or voltage constraints (Table D-6) were identified.

3.3 Network Upgrades Identified in MISO ERIS Analysis for 2024 Summer Peak Scenario

Based on the MISO 2024 summer peak steady state analyses, no thermal constraints or voltage constraints were identified; no MISO thermal NUs or reactive power NUs are required for DPP 2018 April West projects.

Section

4

Summer Shoulder Steady-State Analysis

Summer shoulder steady-state analysis was performed in summer shoulder scenario to identify thermal and voltage upgrades required for interconnecting the generating facilities in the DPP 2018 April West Area group to the transmission system.

4.1 Study Procedure

Summer shoulder power flow case was created in the procedure as described in Section 2.1. Due to post-contingent low voltage violations (<0.9 p.u.) at a few critical 345 kV buses in the initial power flow case, steady-state analysis was performed in the following three-step procedure:

- Step 1: Non-linear (AC) contingency analysis (Stage-1 ACCC) was performed for several critical contingencies to identify low voltage violations.
- Step 2: Based on the identified voltage violations in the Stage-1 ACCC, project justification analysis was performed to determine Network Upgrades (NUs) required for interconnection of DPP West projects.
- Step 3: The justified NUs were added to the models. Stage-2 ACCC was performed to identify any remaining thermal and voltage constraints.

4.2 Step 1 – Stage-1 ACCC Analysis

AC contingency analysis was performed for the below critical contingencies identified in the Phase 2 study.

- Wilmarth-Sheas Lake 345 kV
- Helena-Sheas Lake 345 kV
- Helena-Scott County 345 kV
- Brookings Co-Astoria 345 kV
- Mandan-J302 POI 230 kV
- Wishek-J302 POI 230 kV

Low voltage violations (below or around 0.9 p.u.) were identified at the following 200 kV and above critical buses:

- Cedar Mountain 345 kV
- J901 POI 345 kV
- J1106 POI 345 kV
- McLeod 230 kV

- Panther 230 kV
- Lyon County 345 kV
- Helena 345 kV
- Hazel Creek 345 kV
- Sheas Lake 345 kV

4.3 Step 2 – NUs Justification Analysis

Since voltages at Cedar Mountain 345 kV buses are the lowest (<0.89 p.u.) under the tested critical contingencies, 2×75 MVar switched capacitor at GRE-CDRMTH23 345 kV bus (bus #: 615647) was tested for mitigation. It was concluded that this 2×75 MVAR switched capacitor at Cedar Mountain 345 kV bus can effectively mitigate the identified low voltage violations at several critical buses. Detailed NU justification results are in Appendix E.1.

4.4 Step 3 – Stage-2 ACCC Analysis

The 2×75 MVar switched capacitor was added at Cedar Mountain 345 kV bus to create Stage-2 models. AC contingency analysis was performed in the Stage-2 models to identify any remaining thermal and voltage constraints.

4.4.1 Stage-2 ACCC Analysis Results for Summer Shoulder Condition

4.4.1.1 System Intact Conditions

For NERC category P0 (system intact) conditions, thermal constraints are listed in Table E-2. No voltage constraints were identified (Table E-3).

4.4.1.2 Post Contingency Conditions

The results in this Section are for analysis of conditions following NERC Category P1-P7 contingencies. All NERC Category P1 contingencies were converged.

For P1 contingencies, thermal constraints are listed in Table E-4. No voltage constraints were identified (Table E-5).

Two category P2-P7 contingencies (Table E-8) were not converged in both the benchmark and study cases. No mitigation plan is required for the study projects for these non-converged contingencies.

For the non-converged contingencies in Table E-8, DC contingency analysis was performed to get the dc thermal results. The dc thermal results for non-converged contingencies are listed in Table E-9.

For P2-P7 contingencies, thermal constraints are listed in Table E-6. No voltage constraints were identified (Table E-7).

4.4.2 Summer Shoulder Worst Thermal Constraints in the Stage-2 ACCC

Table 4-1 lists worst thermal constraints and Network Upgrades identified in the Stage-2 ACCC for 2024 summer shoulder scenario.

Table 4-1: 2024 Summer Shoulder Thermal Constraints, Maximum Screened Loading, Stage-2 ACCC

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Mitigation
				(MVA)	(%)			
J982	J748 POI-Raun 345 kV	864.0	MEC	944.7	109.3	CEII Redacted	P1	MEC: Structure replacements. New rating projected to be 970 MVA.
J982	J748 POI-Raun 345 kV	864.0	MEC	946.7	109.6	CEII Redacted	P2-P7	MEC: Structure replacements. New rating projected to be 970 MVA.
J982	Lakefield-Huntley 345 kV	1195.0	ITCM	1215.7	101.7	CEII Redacted	P2-P7	Lakefield to Huntley 345 kV is rated 1793 MVA summer
J982	Lakefield-Huntley 345 kV	1195.0	ITCM	1215.7	101.7	CEII Redacted	P2-P7	Lakefield to Huntley 345 kV is rated 1793 MVA summer
J975	Sheyenne-Lake Park 230 kV	301.0	XEL MPC OTP	327.9	108.9	CEII Redacted	P0	OTP: Structure replacement. \$600,000. 2017 Aug. Ph3 OTP LPC upgrade
J975,J1040	Sheyenne-Lake Park 230 kV	301.0	XEL MPC OTP	332.8	110.6	CEII Redacted	P1	OTP: Structure replacement. \$600,000. 2017 Aug. Ph3 OTP LPC upgrade

Summer Shoulder Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Mitigation
				(MVA)	(%)			
J975	Sheyenne-Mapleton 115 kV	262.9	XEL OTP	271.3	103.2	CEII Redacted	P1	Sheyenne-Mapleton 115 kV Rebuild. \$17.6M. 2017 Aug. Ph3 upgrade
J975	Sheyenne-Mapleton 115 kV	262.9	XEL OTP	277.0	105.3	CEII Redacted	P2-P7	Sheyenne-Mapleton 115 kV Rebuild. \$17.6M. 2017 Aug. Ph3 upgrade
J982	Rutland-Fox Lake 161 kV	195.0	ITCM SMMPA	218.3	111.9	CEII Redacted	P1	ITCM: Line is currently sag limited at 185 F. Mitigation required at 5 spans of line. Rating will increase to 219 MVA. \$1.1M
J975	CSLTNET-Mapleton 115 kV	263.0	OTP	277.4	105.5	CEII Redacted	P1	Replace switch at Mapleton
J975	CSLTNET-Mapleton 115 kV	263.0	OTP	283.0	107.6	CEII Redacted	P2-P7	Replace switch at Mapleton
J975	Buffalo 345-230-41.6 kV xfmr #2	140.0	OTP	151.0	107.8	CEII Redacted	P2-P7	Replace Buffalo transformer #2 with larger unit.
J1040	Wahpeton-Fergus Falls 230 kV	379.1	OTP MRES	381.8	100.7	CEII Redacted	P0	OTP: Structure replacement. \$800,000. 2017 Aug. Ph3 OTP LPC upgrade

Summer Shoulder Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Mitigation
				(MVA)	(%)			
J975,J1040	Wahpeton-Fergus Falls 230 kV	379.1	OTP MRES	383.4	101.1	CEII Redacted	P1	OTP: Structure replacement. \$800,000. 2017 Aug. Ph3 OTP LPC upgrade
J975	Audubon-Lake Park 230 kV	294.0	OTP	320.0	108.8	CEII Redacted	P0	OTP: Structure replacement. \$100,000. 2017 Aug. Ph3 OTP LPC upgrade
J975,J1040	Audubon-Lake Park 230 kV	294.0	OTP	324.8	110.5	CEII Redacted	P1	OTP: Structure replacement. \$100,000. 2017 Aug. Ph3 OTP LPC upgrade
J1040	Ellendale-Aberdeen 115 kV	128.0	MDU NWE	131.6	102.8	CEII Redacted	P1	Ellendale-Aberdeen Jct 115 kV Uprate.
J1040	Ellendale-Aberdeen 115 kV	128.0	MDU NWE	131.7	102.9	CEII Redacted	P2-P7	Ellendale-Aberdeen Jct 115 kV Uprate.
J1040	J302 POI-Mandan 230 kV	383.0	MDU	418.5	109.3	CEII Redacted	P1	Uprate Mandan-J302 POI 230 kV line (new rating: 471/471 MVA [N/E]). J302/503 upgrade
J1040	J302 POI-Mandan 230 kV	383.0	MDU	561.1	146.5	CEII Redacted	P2-P7	Add a breaker at Merricourt which will eliminate tripping of both 230 kV lines for the "P23:230:MDU::MERRICOURT::2722" contingency.
J1040	Merricourt-Wishek 230 kV	343.0	MDU	412.4	120.2	CEII Redacted	P1	Rebuild/reconductor line with 795 ACSS (new rating: 478/511 MVA [N/E]).

Summer Shoulder Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Mitigation
				(MVA)	(%)			
J1040	Merricourt-Wishek 230 kV	343.0	MDU	412.4	120.2	CEII Redacted	P2-P7	Rebuild/reconductor line with 795 ACSS (new rating: 478/511 MVA [N/E]).
J967,J1072,J1124,J1181	Wabaco-Rochester 161 kV	221.1	DPC	265.3	120.0	CEII Redacted	P1	A third party will fund the Rochester-Wabaco 161kV rebuild. \$11M. MTEP Appendix A project
J967,J1181	Wabaco-Rochester 161 kV	221.1	DPC	329.0	148.8	CEII Redacted	P2-P7	A third party will fund the Rochester-Wabaco 161kV rebuild. \$11M. MTEP Appendix A project
J967,J1181	Wabaco-Alma 161 kV	291.0	DPC	313.9	107.9	CEII Redacted	P2-P7	A third party will upgrade the first section of the Wabaco-Alma 161kV line. \$6.3M

4.5 Network Upgrades Identified in MISO ERIIS Analysis for 2024 Summer Shoulder Scenario

Based on the MISO 2024 summer shoulder steady state analyses, the reactive power NU required for mitigating low voltages at Cedar Mountain 345 kV buses under several critical contingencies are listed in Table 4-2. Thermal NUs and cost are listed in Table 4-3.

Table 4-2: Reactive Power Network Upgrades Required for Mitigating Low Voltages at Cedar Mountain 345 kV Buses

NUs	Cost (\$)
2x75 MVar switched capacitor at Cedar Mountain 345 kV	\$3,000,000

Table 4-3: Thermal NUs

Constraint	Owner	Mitigation	Cost (\$)
J748 POI-Raun 345 kV	MEC	MEC: Structure replacements. New rating projected to be 970 MVA.	\$1,500,000
Lakefield-Huntley 345 kV	ITCM	Lakefield to Huntley 345 kV is rated 1793 MVA summer	\$0
Sheyenne-Lake Park 230 kV	XEL MPC OTP	OTP: Structure replacement. \$600,000. 2017 Aug. Ph3 OTP LPC upgrade	\$0 ¹
Sheyenne-Mapleton 115 kV	XEL OTP	Sheyenne-Mapleton 115 kV Rebuild. \$17.6M. 2017 Aug. Ph3 upgrade	\$0 ¹
Rutland-Fox Lake 161 kV	ITCM SMPMA	ITCM: Line is currently sag limited at 185 F. Mitigation required at 5 spans of line. Rating will increase to 219 MVA. \$1.1M	\$1,100,000
CSLTNET-Mapleton 115 kV	OTP	Replace switch at Mapleton	\$200,000
Buffalo 345-230-41.6 kV xfmr #2	OTP	Replace Buffalo transformer #2 with larger unit.	\$3,000,000
Wahpeton-Fergus Falls 230 kV	OTP MRES	OTP: Structure replacement. \$800,000. 2017 Aug. Ph3 OTP LPC upgrade	\$0 ¹
Audubon-Lake Park 230 kV	OTP	OTP: Structure replacement. \$100,000. 2017 Aug. Ph3 OTP LPC upgrade	\$0 ¹
Ellendale-Aberdeen 115 kV	MDU NWE	Ellendale-Aberdeen Jct 115 kV Uprate.	\$60,000
J302 POI-Mandan 230 kV	MDU	Uprate Mandan-J302 POI 230 kV line (new rating: 471/471 MVA [N/E]). J302/503 upgrade	\$0
J302 POI-Mandan 230 kV (P2-3)	MDU	Add a breaker at Merricourt which will eliminate tripping of both 230 kV lines for the "P23:230:MDU::MERRICOURT::2722" contingency.	\$1,500,000
Merricourt-Wishek 230 kV	MDU	Rebuild/reconductor line with 795 ACSS (new rating: 478/511 MVA [N/E]).	\$9,000,000

Constraint	Owner	Mitigation	Cost (\$)
Wabaco-Rochester 161 kV	DPC	A third party will fund the Rochester-Wabaco 161kV rebuild. \$11M. MTEP Appendix A project	\$0 ²
Wabaco-Alma 161 kV	DPC	A third party will upgrade the first section of the Wabaco-Alma 161kV line. \$6.3M	\$0 ²

Note 1: Costs of Network Upgrades required in DPP 2017 Aug West Ph3 are currently assigned to projects in DPP 2017 Aug. Phase 3 cycle. The NUs cost may be assigned to DPP 2018 Apr. projects if projects in DPP 2017 Aug. are withdrawn.

Note 2: If the agreement with the third party falls through, DPP 2018 Apr. projects will be responsible for the upgrades to Rochester-Wabaco and Wabaco -Alma lines.

Section

5

Local Planning Criteria Analysis

Local Planning Criteria (LPC) analyses were performed to identify additional constraints per Transmission Owning Companies' LPC.

5.1 OTP Local Planning Criteria Analysis

J975 is to be interconnected in Otter Tail Power (OTP) transmission system. In addition to MISO's standard DPP analysis, OTP determined that J975 is required for OTP LPC study. The OTP LPC analysis consisted of steady-state contingency analysis for summer shoulder, summer peak, and Light Load No Wind conditions.

Three additional scenarios were analyzed in the OTP LPC study:

1. OTP LPC Summer Peak (SPK)
2. OTP LPC Summer Shoulder (SH)
3. OTP LPC Light Load No Wind (LLNW)

Siemens PTI performed the local planning criteria analysis based on OTP's LPC. The OTP local planning criteria analysis details can be found in Appendix F.1.

5.1.1 Additional Network Upgrades Identified in OTP LPC Analysis

No thermal or voltage constraints were identified in the OTP LPC analysis. No additional Network Upgrades are required in the OTP LPC study.

5.2 MDU Local Planning Criteria Analysis

Montana-Dakota Utilities Company (MDU) determined that J1040 has more than 20% DF on the outlets of Tatanka, Foxtail, J436/J437/J488, J302/J503, G359 (Merricourt). In addition to MISO's standard DPP analysis, MDU determined that J1040 is required for MDU LPC study per Section 3.2 of the MDU Local Planning Criteria. The MDU LPC analysis consisted of steady-state contingency analysis and stability analysis for summer shoulder condition.

One additional scenario was analyzed in the MDU LPC study:

- a. MDU LPC Summer Shoulder (SH)

Siemens PTI performed the local planning criteria analysis based on MDU's LPC. The MDU local planning criteria analysis details can be found in Appendix F.2.

5.2.1 Additional Network Upgrades Identified in MDU LPC Analysis

Additional Network Upgrades required in the MDU LPC study are listed in Table 5-1.

Table 5-1: Additional Network Upgrades for Constraints Identified in MDU LPC Analysis

Constraint	Owner	Mitigation	Cost (\$)
Ellendale-Aberdeen 115 kV	MDU NWE	MDU: NWPS owns the line. Minor substation upgrades at MDU's Ellendale 115kV terminal. New rating for MDU's terminal is 192/192 [SN/SE].	\$250,000
J302 POI-Mandan 230 kV	MDU	Minor substation upgrades. New rating is 608/608 [SN/SE].	\$50,000
J302 POI-Mandan 230 kV (P2-3)	MDU	MISO NU. Add a breaker at Merricourt which will eliminate tripping of both 230 kV lines for the "P23:230:MDU::MERRICOURT::2722" contingency.	\$0 ¹
Wishek-J302 POI 230 kV (P2-3)	MDU	MISO NU. Add a breaker at Merricourt which will eliminate tripping of both 230 kV lines for the "P23:230:MDU::MERRICOURT::2722" contingency.	\$0 ¹

Note 1: This is the NU required in MISO steady state analysis.

Affected System Steady-State Analysis

Steady state analyses were performed to identify constraints in affected systems.

6.1 Affected System Analysis for CIPCO Company

Per CIPCO Affected System Planning Criteria, a CIPCO transmission facility is a constraint if it satisfies all three of the following conditions:

1. the branch is loaded above its applicable normal or emergency rating for the post-change case, and
2. the generator has a larger than 3% DF on the overloaded facility under post contingent condition or 5% DF under system intact condition, and
3. the loading increase of the overloaded facility is greater than 1 MVA compared with that in the pre-change case under system intact or contingency conditions.

AC contingency analysis was performed for this CIPCO affected system analysis, using the following benchmark and study cases:

- Summer peak benchmark and study cases
- Summer shoulder benchmark and study cases

All NERC category P0-P7 contingencies described in Section 2.2 were simulated. The CIPCO affected system was monitored.

CIPCO affected system analysis thermal results are listed in Appendix G.1. There are no CIPCO thermal constraints for both summer shoulder and summer peak conditions.

6.2 MPC Affected System Analysis

The MPC affected system analysis details can be found in Appendix G.2.

6.2.1 Study Summary

Minnkota Power Cooperative (MPC) performed an Affected System Analysis (ASA) to determine impacts of generators in the MISO DPP 2018 April Phase 3 study cycle on MPC facilities and any network upgrades required to mitigate those impacts. Steady-state power flow and contingency analysis, and dynamic stability analysis were performed for two DPP generating facilities.

- J975, ERIS, 150 MW wind at Buffalo 115 kV substation
- J1040, NRIS, 250 MW wind at Wishek Junction 230 kV substation

6.2.2 Network Upgrades

The Network Upgrades required to mitigate constraints identified in the Minnkota ASA are listed in Table 6-1 through Table 6-2. Costs are planning level estimates and subject to revision in the facility studies.

Table 6-1. Minnkota Network Upgrades Allocated to DPP 2018 Projects

Constraint	Owner	Highest Loading (MVA)	Mitigation	Cost (\$)	Generators
Wilton-Winger 230	OTP MPC	380	MPC: Resag to 90°C. OTP: Mitigation not required for ASA	\$500,000	J975, J1040
Drayton 230-115 kV xfmr	MPC	166	Replace transformer.	\$2,100,000	J1040
Prairie-Walle 230	MPC	414	Resag to 95°C.	\$200,000	J1040
Winger-Walle 230	MPC	389	Resag to 85°C.	\$500,000	J1040
MPC Steady State Voltage	-	-	Additional 1x75 MVar MSC at Bison 345 (601067)	\$1,500,000	J975, J1040
MPC Steady State Voltage	-	-	2x40 MVar MSC at Maple River 230 (657754)	\$2,000,000	J975, J1040
Transient Instability	-	-	+200 MVar SVC at Winger 230 (657758)	\$30,000,000	J975, J1040
Transient Instability	-	-	+50 MVar SVC at Frontier 230 (657750)	\$15,000,000	J975, J1040
Transient Instability	-	-	2x30 MVar MSC at Frontier 230 (657750)	\$2,000,000	J975, J1040
Total				\$53,800,000	

Table 6-2: Minnkota Network Upgrades Allocated to DPP 2017 Aug Projects

Constraint	Owner	Highest Loading (MVA)	Mitigation	Generators
Grand Forks-Falconer 115	MPC	278	DPP 2017 Aug ASA NU expected to replace conductor, CB, switches and CTs.	J1040
Falconer-Oslo 115	MPC	205	DPP 2017 Aug ASA NU expected to reductor with 954 ACSR Rail and rebuild Oslo bus.	J1040
Voltage collapse	MPC	-	Install 2x75 MVar MSC at Bison 345 (601067)	J975, J1040
Voltage collapse	MPC	-	Install 2x30 MVar MSC at Winger 230 (657758)	J975, J1040

6.3 PJM Affected System Analysis

The PJM affected system analysis details (dated 05/28/2021) can be found in Appendix G.3.

6.3.1 Study Results

6.3.1.1 Overload on Nelson;B- Electric JCT;R 345 kV line

The upgrade will be to re-conductor the line, station conductor work and upgrade 2-disconnect switches. A preliminary estimate for the upgrade is \$36.2 M.

These queue projects contribute to the constraint: J1084, J1131, J963, J1181, J1135, J1000, J1050, J1174, J1175, J967, J1072, J1110, and J982.

The cost allocation is as follows:

Queue	MW Contribution	Percentage of Cost	Cost (\$36.2M)
AE1-114	20.9	22.57%	\$8.17
J982	20.8	22.46%	\$8.13
J1084	31.7	34.23%	\$12.39
J1181	19.2	20.73%	\$7.51

6.3.2 Study Summary

Multiple projects in the MISO DPP 2018 April West Area group contribute loading to the overloads in the PJM system. Some of these projects are responsible for the cost of Network Upgrades per PJM cost allocation rules.

6.4 AECI Affected System Analysis

The AECI affected system analysis details (dated 5/14/2021) can be found in Appendix G.4.

6.4.1 Study Results

Associated Electric Cooperative Inc. (AECI), through coordination with the Midcontinent Independent System Operator (MISO), has updated the analysis for generator interconnection requests (GIRs) within the DPP-2018-APR Study Cycle (the “Study Cycle”) for an Affected System Study (AFS) evaluation on the AECI transmission system.

Steady state analysis was performed to confirm the reliability impacts on the AECI system under a variety of system conditions and outages. AECI’s transmission system must be capable of operating within the applicable normal ratings, emergency ratings, and voltage limits of AECI planning criteria.

Steady state analysis results showed three (3) new thermal violations on the AECI system reported due to the addition of the Study Cycle projects.

AECI developed non-binding, good faith estimates of the timing and cost estimates for upgrades needed as a result of the addition of the Study Cycle projects, as shown in Table 6-3. The associated cost allocation of the network upgrades to each of the Study Cycle projects is provided in Table 6-4.

Generation projects in DPP 2018 April West Phase 3 study cycle are not responsible for the cost of Network Upgrades identified in the AECI Affected System Study.

Table 6-3: Network Upgrade Costs

ID	Option / Description	Cost*	Year In Service
NU01	Rebuild 2.12-mile-long segment of Green Forest to Township 69 kV line to 336 ACSR	\$2,895,000	TBD
*2020\$, includes engineering and contingencies		Total Cost:	\$2,895,000

Table 6-4: Network Upgrade Cost Allocation

Project	NU01	Total Cost
J1007	\$65,446	\$65,446
J1034	\$1,396,647	\$1,396,647
J1060	\$159,245	\$159,245
J1087	\$721,580	\$721,580
J1107	\$491,670	\$491,670
J1125	\$60,412	\$60,412
Total Cost	\$2,895,000	\$2,895,000

6.5 SPP Affected System AC Contingency Analysis

Southwest Power Pool (SPP) conducted an Affected System Impact Study (ASIS) to evaluate potential impacts to the SPP Transmission System related to the interconnection of generators on the Mid-Continent Independent System Operation (MISO) Transmission System.

A steady-state thermal and voltage analysis as well as Transfer Distribution Factor analysis was performed to determine the impact the MISO GIRs have on the SPP system.

ERIS constraints identified in the SPP affected system are listed in Table 6-5.

NRIS constraints identified in the SPP affected system are listed in Table 6-6.

Cost allocation of SPP Network Upgrades are listed in Table 6-7.

The SPP affected system analysis results (06/02/2021) for this study are in Appendix G.5.

Table 6-5: SPP ERIS Constraints

Monitored Facility	Mitigation
Split Rock to Sioux City 230 kV	Upgrade Terminal Equipment at Sioux Falls 230 kV
Raun to S3451 345 kV	Rebuild Raun to S3451 345 kV Circuit
Split Rock to Sioux City 230 kV	Upgrade Terminal Equipment at Split Rock 230 kV

Table 6-6: SPP NRIS Constraints

Monitored Facility	Mitigation
Grand Forks to Falconer 115 kV	Upgrade Terminal Equipment at Grand Forks 115 kV

Table 6-7: SPP Network Upgrades Cost Allocation

Interconnection Request	ERIS MW	NRIS MW	ERIS	NRIS	Total	ERIS Total	NRIS Total	Total
J1025	290	290	\$0	\$0	\$0	\$85,220,871	\$90,000	\$85,310,871
J1026	380	265	\$0	\$0	\$0			
J1034	225	225	\$0	\$0	\$0			
J1040	250	250	\$11,012,024	\$90,000	\$11,102,024			
J1045	20	20	\$1,426,716	\$0	\$1,426,716			
J1050	225	225	\$0	\$0	\$0			
J1072	150	150	\$4,631,343	\$0	\$4,631,343			
J1087	200	200	\$0	\$0	\$0			
J1092	100	100	\$3,476,653	\$0	\$3,476,653			
J1107	200	200	\$0	\$0	\$0			
J1110	100	100	\$3,383,313	\$0	\$3,383,313			
J1124	100	100	\$3,401,667	\$0	\$3,401,667			
J1132	50	0	\$0	\$0	\$0			
J1145	250	250	\$0	\$0	\$0			
J1164	80	0	\$6,320,652	\$0	\$6,320,652			
J1169	50	0	\$4,420,468	\$0	\$4,420,468			
J1174	165	165	\$6,897,617	\$0	\$6,897,617			
J1175	165	0	\$6,897,617	\$0	\$6,897,617			
J1181	200	0	\$4,668,050	\$0	\$4,668,050			
J1182	250	250	\$0	\$0	\$0			
J953	2	2	\$0	\$0	\$0			
J954	1.4	1.4	\$0	\$0	\$0			
J956	200	200	\$0	\$0	\$0			
J967	150	0	\$4,631,343	\$0	\$4,631,343			
J975	150	0	\$6,336,633	\$0	\$6,336,633			
J976	300	300	\$0	\$0	\$0			
J982	300	0	\$17,716,774	\$0	\$17,716,774			
J987	100	100	\$0	\$0	\$0			
J994	100	100	\$0	\$0	\$0			

Stability Analysis

Stability analysis was performed to evaluate the transient stability and impact on the region of the generating facilities in the DPP 2018 April West study cycle.

7.1 Procedure

7.1.1 Computer Programs

Stability analysis was performed using TSAT revision 19.0.

7.1.2 Study Methodology

A stability package representing 2024 summer peak (PK) and summer shoulder (SH) conditions with generating facilities in the DPP 2018 April cycle was created from the MTEP19 stability package. Disturbances were simulated to evaluate the transient stability and impact on the region of the generating facilities. MISO transient stability criteria and local TOs' planning criteria specified in MTEP19 were adopted for checking stability violations.

7.2 Case Development

7.2.1 Summer Peak (PK) Stability Model

Summer peak stability model is the same as the summer peak steady state model. The model does not have the fictitious SVCs in SPP (Table 2-1).

7.2.2 Summer Shoulder (SH) Stability Model

Summer shoulder stability model was created from the summer shoulder steady state model (Section 2.1). The summer shoulder stability model includes the following Network Upgrades which are also required by summer shoulder steady state model:

- 200 Mvar fictitious SVC at Mingo 345 kV bus in SPP (Table 2-1).
- 2×75 MVar switched capacitor at Cedar Mountain 345 kV bus (bus #: 615647).

7.3 Disturbance Criteria

The stability simulations performed as part of this study considered all the regional and local contingencies listed in Table 7-1. Regional contingencies with pre-defined switching sequences were selected from the MISO MTEP19 study; switching sequences for local contingencies were developed based on the generic clearing times shown in Table 7-2. The admittance for local single line-to-ground (SLG) faults were estimated by assuming that the

Thevenin impedance of the positive, negative and zero sequence networks at the fault point are equal.

Table 7-1: Regional and Local Disturbance Descriptions

CEII Redacted

Table 7-2: Generic Clearing Time Assumption

Voltage Level (kV)	Primary Clearing Time (cycle)	Backup Clearing Time (cycle)
345 kV	4	11
230 kV	5	13
161/138 kV	6	18
115 kV	6	20
69 kV	8	24

7.4 Performance Criteria

MISO transient stability criteria and local TOs' planning criteria specified in MTEP19 were adopted. All generators must mitigate the stability constraints to obtain any type of Interconnection Service.

7.5 Summer Peak Stability Results

The contingencies listed in Table 7-1 were simulated using the summer peak stability case without any steady state ERIIS Network Upgrades identified in DPP 2018 April Phase 3.

Appendix H.1.2 contains plots of generator rotor angles, generator power output, and bus voltages for each simulation. Simulations were performed with a 0.5 seconds steady-state run followed by the appropriate disturbance. Simulations were run for a 10-second duration.

Summer peak stability study results summary is in Appendix H.1.1, Table H-1.

The following stability related issues were identified in the summer peak stability study.

7.5.1 Transient Voltage Rise at Arnold 161 kV Bus

Under the fault of "2298_w_mec_p55" (Table 7-3), transient voltage at Arnold 161 kV bus was above 1.04 p.u. for 1.308 second (>1.0 second threshold). This transient voltage rise can be resolved by resetting generator's scheduled voltage, turning off capacitor at Arnold 161 kV bus, or adjusting transformer tap position.

Table 7-3: Transient Voltage Rise at Arnold 161 kV Bus

CEII Redacted

7.5.2 Transient High Voltage at Sub T HSK 345 kV Bus

Under faults of “0765_w_mec_p42” and “2282_w_mec_P42” (Table 7-4), transient voltage at Sub T HSK 345 kV bus is above 1.05 p.u. for 9.266 second (>5.01 second threshold). This transient high voltage can be resolved by switching on the reactor at the Sub T HSK 345 kV bus.

Table 7-4: Transient High Voltage at Sub T HSK 345 kV Bus

CEII Redacted

7.5.3 Stability Network Upgrades Identified in Summer Peak

In summary, there are no stability Network Upgrades identified in summer peak stability study.

7.6 Summer Shoulder Stability Results

The contingencies listed in Table 7-1 were simulated using the summer shoulder stability case described in Section 7.2.2.

Appendix H.2.2 contains plots of generator rotor angles, generator power output, and bus voltages for each simulation. Simulations were performed with a 0.5 seconds steady-state run followed by the appropriate disturbance. Simulations were run for a 10-second duration.

Summer shoulder stability study results summary is in Appendix H.2.1, Table H-2.

The following stability related issues were identified in the summer shoulder stability study.

7.6.1 Voltage Collapse under Two Faults

Under two faults listed in Table 7-5, transient instability and voltage collapse was identified in Alexandria 345 kV area in both benchmark and study cases.

Table 7-5: Voltage Collapse under Two Faults in Both Benchmark and Study Cases

CEII Redacted

With addition of Network Upgrades listed in Table 7-6, the identified transient instability and voltage collapse in the benchmark case can be mitigated. But there are still transient voltage violations at many buses, such as: Audubon, Blue Lake, Erie Jct, Panther, Riverview, McLeod, Lake Park, Paynes, Quarry, Riverton areas, etc.

Table 7-6: Additional Stability Network Upgrades for Mitigating Voltage Collapse

Network Upgrades	Owner	Cost (\$)	Comments
Build Alexandria - Twin Brooks 345 kV line	OTP MRES	\$242,400,000	Only required in benchmark case ¹
Four 50 MVAR capacitors at Alexandria 345 kV	MRES	\$16,000,000	Required in both benchmark and study cases

Note 1: Alexandria -Twin Brooks 345 kV line is only required for the benchmark case.

With addition of 4×50 Mvar switched capacitor at Alexandria 345 kV bus (without Alexandria – Twin Brooks 345 kV line), the identified transient instability and voltage collapse in the study case can be mitigated. But there are still transient voltage violations with less or similar severity as those identified in the benchmark case (with NUs in Table 7-6). If Alexandria-Twin Brooks 345 kV line is also added to the study case, all transient voltage violations will be mitigated in the study case.

Since the Network Upgrades listed in Table 7-6 are required for the benchmark case, DPP 2018 April West generation projects are not responsible for these Network Upgrades.

7.6.2 Transient Voltage Rise/Drop at Arnold 161 kV Bus

Under multiple faults listed in Table 7-7, transient voltage at Arnold 161 kV bus was above 1.04 p.u. for more than 1.0 second. Under the fault of “1174_x_ce_p12”, transient voltage at Arnold 161 kV bus dropped to below 0.99 p.u. for more than 1.0 second. The transient voltage rise/drop can be resolved by resetting generator’s scheduled voltage, turning off capacitor at Arnold 161 kV bus, or adjusting transformer tap position.

Table 7-7: Transient Voltage Rise/Drop at Arnold 161 kV Bus

CEII Redacted

7.6.3 Transient High Voltage at Kossuth, Irvington 345 kV Buses

Under fault of “2283_w_mec_P42” (Table 7-8), transient voltages at Kossuth and Irvington 345 kV buses are above 1.05 p.u. for more than 5.01 second. The transient high voltages can be resolved by switching on the reactor at the Kossuth 345 kV bus.

Table 7-8: Transient High Voltage at Kossuth, Irvington 345 kV Buses

CEII Redacted

7.6.4 Stability Network Upgrades Identified in Summer Shoulder

In summary, additional stability Network Upgrades required in summer shoulder stability study are listed in Table 7-9.

Table 7-9: Additional Stability Network Upgrades Required in Summer Shoulder Study

Network Upgrades	Owner	Cost (\$)	Comments
Build Alexandria - Twin Brooks 345 kV line	OTP MRES	\$242,400,000	Only required in benchmark case ¹
Four 50 MVAR capacitors at Alexandria 345 kV	MRES	\$16,000,000	Required in both benchmark and study cases

Note 1: Alexandria -Twin Brooks 345 kV line is only required for the benchmark case.

7.7 Additional Network Upgrades Identified in Stability Analysis

Additional Network Upgrades required in the DPP 2018 April Phase 3 stability analysis are listed in Table 7-9. Since the Network Upgrades listed in Table 7-9 are required for the

benchmark case, and voltage collapse in the study case is less severe than in the benchmark case, DPP 2018 April West generation projects are not responsible for these Network Upgrades.

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MWEX Voltage Stability Study

ATC performed steady state voltage stability analysis. Voltage stability analysis is required to determine if the initial conditions of the DPP system models under study are in a stable state as defined by Power-Voltage (PV) curves of the Minnesota Wisconsin Export Interface (MWEX) for the worst contingency.

As shown in Table 8-1, the Pre-DPP scenario in the 2024SH case is not voltage stable because the initial condition does not converge with two contingencies. The Post-DPP scenario in the 2024SH case alleviates this criteria violation, is voltage stable, and meets all other criteria. Therefore, Network Upgrades related to MWEX voltage stability will not be assigned to the Interconnection Customers.

The MWEX voltage stability study details can be found in Appendix I.

Table 8-1: MWEX Margins to Collapse in the 2024SH Cases

	Real Power Flow (MW)							
	AHD-PAR ²	MWEX			Post-Contingent Margin to Nose ³		MWEX Control Point	
Case	N-0 Initial Condition	N-0 I.C. ⁴	N-1 I.C. ⁴	N-1 Nose	(MW)	(%)	(MW)	Notes
Bench	403	1011	N/A - Bench case does not converge					Voltage Unstable
Study	405	994	503	752	249	33.2	1423	Voltage Stable, Sufficient Margin

Notes:

- Based on 50 MW transfer intervals.
- As described in the active MWEX Operating Guide, the AHD-SLK interface is a single element Power Transfer Distribution Factor (PTDF) interface measured at the Minnesota Power 230 kV side of the Arrowhead 230 kV phase shifter. Due to the addition of the Superior substation, the AHD-SLK interface has here been re-named to AHD-PAR to avoid confusion.
- Post-Contingent Margin to Nose is defined as:
 - "Post-Contingent Margin to Nose (MW)" = "MWEX N-1 Nose" – "N-1 Initial Condition"
 - "Post-Contingent Margin to Nose (%)" = "Post-Contingent Margin to Nose (MW)" / "MWEX N-1 Nose"
- Initial Condition flows were measured in the base cases with an intact system and the worst contingency, plus operation of various control systems as needed with all transformer taps, switched shunts, and PARs locked.
- ATC Planning Criteria requires:

- a. The N-0 Initial Condition must be less than or equal to the Control Point (CP). The CP is 98% of the System Operating Limit (SOL). The SOL is the pre-contingent flow that results in a post-contingent flow that is 95% of the post-Contingency P-V curve nose.
- b. $V_{nose} < V_{min}$.

Short Circuit Analysis

Siemens PTI and several transmission owning companies performed short circuit analysis for the DPP 2018 April West study cycle projects.

9.1 J967 & J1072 Short Circuit Study

The J967 & J1072 short circuit study was performed by Siemens PTI. The study results show that the 3PH fault current is 15,468 A (increased by 706 A) and SLG fault current is 13,582 A (increased by 1,527 A) at the Adams 345 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J967 and J1072 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix J.1.

9.2 J975 Short Circuit Study

The J975 short circuit study was performed by OTP. Based on the short circuit analysis performed, the fault current ratings of the Transmission Owner's equipment in the area are not exceeded and there are no upgrades required. With the proposed projects additions, the fault currents are roughly 11.8 kA at the Buffalo 115 kV bus. There does not appear to be any short circuit related upgrades needed for the projects. The Transmission Owner did not evaluate any impacts on the fault-current levels at substations owned by other Transmission Owners.

Study details can be found in Appendix J.2.

9.3 J982 Short Circuit Study

The J982 short circuit study was performed by Siemens PTI. The study results show that the 3PH fault current is 11,657 A (increased by 1,085 A) and SLG fault current is 10,104 A (increased by 2,018 A) at the J982 POI 345 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J982 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix J.3.

9.4 J1040 Short Circuit Study

The J1040 short circuit study was performed by Siemens PTI. The study results show that the 3PH fault current is 6,276 A (increased by 1,119 A) and SLG fault current is 6,188 A (increased by 1,689 A) at the Wishek 230 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J1040 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix J.4.

9.5 J1045 Short Circuit Study

The J1045 short circuit study was performed by Siemens PTI. The study results show that the 3PH fault current is 10,632 A (increased by 162 A) and SLG fault current is 12,005 A (increased by 162 A) at the J874 substation (J874SUB) 115 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J1045 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix J.5.

9.6 J1050 Short Circuit Study

The J1050 short circuit study was performed by ITCM. Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J1050. Equipment not owned by ITC Midwest was not evaluated for interrupting capability.

Study details can be found in Appendix J.6.

9.7 J1084 Short Circuit Study

The J1084 short circuit study was performed by ITCM. Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J1084. Equipment not owned by ITC Midwest was not evaluated for interrupting capability.

Study details can be found in Appendix J.7.

9.8 J1092 Short Circuit Study

The J1092 short circuit study was performed by Siemens PTI. The study results show that the 3PH fault current is 8,781 A (increased by 615 A) and SLG fault current is 8,748 A (increased by 1,763 A) at the Three Lakes 115 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J1092 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix J.8.

9.9 J1110 Short Circuit Study

The J1110 short circuit study was performed by SMMPA. The study results show that in the study case, 3PH fault current is 13220 Amps (increased by 771 Amps) and SLG fault current is 11182 Amps (increased by 2118 Amps) at the J1128 POI 161 kV bus. Based on the results of the study, SMMPA's and DPC's equipment have adequate interrupting capability to accommodate the interconnection project J1110. Equipment not owned by SMMPA or DPC was not evaluated for interrupting capability.

Study details can be found in Appendix J.9.

9.10 J1124 Short Circuit Study

The J1124 short circuit study was performed by Siemens PTI. The study results show that the 3PH fault current is 17,177 A (increased by 232 A) and SLG fault current is 14,751 A (increased by 823 A) at the Byron 345 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J1124 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix J.10.

9.11 J1131 Short Circuit Study

The J1131 short circuit study was performed by Siemens PTI. The study results show that the 3PH fault current is 29,870 A (increased by 668 A) and SLG fault current is 25,426 A (increased by 290 A) at the Sub 56 161 kV facility. Based on the Transmission Owner's short circuit criteria, interconnection of the J1131 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix J.11.

9.12 J1132 Short Circuit Study

The J1132 short circuit study was performed by ITCM. Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J1132. Equipment not owned by ITC Midwest was not evaluated for interrupting capability.

Study details can be found in Appendix J.12.

9.13 J1135 Short Circuit Study

The J1135 short circuit study was performed by ITCM. Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J1135. Equipment not owned by ITC Midwest was not evaluated for interrupting capability.

Study details can be found in Appendix J.13.

9.14 J1164 Short Circuit Study

The J1164 short circuit study was performed by ITCM. Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J1164. Equipment not owned by ITC Midwest was not evaluated for interrupting capability.

Study details can be found in Appendix J.14.

9.15 J1169 Short Circuit Study

The J1169 short circuit study was performed by Siemens PTI. The study results show that the 3PH fault current is 4,286 A (increased by 423 A) and SLG fault current is 2,970 A (increased by 208 A) at the Grant 115 kV bus. Based on the Transmission Owner's short

circuit criteria, interconnection of the J1169 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix J.15.

9.16 J1174 & J1175 Short Circuit Study

The J1174 and J1175 short circuit study was performed by ITCM. Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Projects J1174 and J1175. Equipment not owned by ITC Midwest was not evaluated for interrupting capability.

Study details can be found in Appendix J.16.

9.17 J1181 Short Circuit Study

The J1181 short circuit study was performed by ITCM. Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J1181. Equipment not owned by ITC Midwest was not evaluated for interrupting capability.

Study details can be found in Appendix J.17.

Deliverability Study

10.1 Project Description

Interconnection requests requesting Network Resource Interconnection Services (NRIS) were considered for deliverability analysis.

10.2 Introduction

Generator interconnection projects must pass Generator Deliverability Study to be granted Network Resource Interconnection Services (NRIS).

If the generator is determined as not fully deliverable, the customer can choose either to change his project to an Energy Resource (ER) project or proceed with the system upgrades that will make the generator fully deliverable.

Generator Deliverability Study ensures that the Network Resources, on an aggregate basis, can meet the MISO aggregate load requirements during system peak condition without getting bottled up. The wind generators are tested at 100 % of their maximum output level which then can be used to meet Resource Adequacy obligations, under Module E, of the MISO Transmission and Energy Market Tariff (TEMT).

10.3 Study Methodology

MISO Generation Deliverability Study method can be found in Appendix C of the MISO Generation Interconnection Business Practices Manual BPM-015-r22.

10.4 2024 Deliverability Study Result

10.4.1 J953

J953 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	1.83 MW (100%)
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10.4.2 J954

J954 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	1.4 MW (100%)
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10.4.3 J963

J963 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	9 MW (100%)
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10.4.4 J1040

J1040 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	250 MW (100%)
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10.4.5 J1045

J1045 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	20 MW (100.0%)
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10.4.6 J1050

J1050 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	225 MW (100%)
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10.4.7 J1072

J1072 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	150 MW (100%)
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10.4.8 J1084

J1084 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	150 MW (100%)
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10.4.9 J1092

J1092 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	100 MW (100%)
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10.4.10 J1110

J1110 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Upgrade Pleasant Valley 345/161 kV Transformer #1	0	0.2219	No		J1110	\$3,000,000	\$3,000,000
Upgrade Pleasant Valley 345/161 kV Transformer #2	100	0.2219	No		J1110	\$3,000,000	\$3,000,000

10.4.11 J1124

J1124 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	100 MW (100%)
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10.4.12 J1131

J1131 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	100 MW (100%)
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10.4.13 J1135

J1135 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	50 MW (100%)
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10.4.14 J1174

J1174 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	165 MW (100%)
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Shared Network Upgrades Analysis

Shared Network Upgrade (SNU) test for Network Upgrades driven by higher queued interconnection project was performed for this System Impact Study.

No SNUs were identified in this study.

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Cost Allocation

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Network Resource Interconnection service as of the draft System Impact Study report date.

12.1 Cost Assumptions for Network Upgrades

The cost estimate for each network upgrade was provided by the corresponding transmission owning company.

12.2 ERIS Network Upgrades Proposed for DPP West Area Projects

Network upgrades for Energy Resource Interconnection Service (ERIS) were identified in the MISO ERIS analysis, LPC analyses, and the affected system analysis. The ERIS network upgrades include voltage network upgrades and thermal network upgrades identified in the MISO steady-state analysis, network upgrades identified in the Local Planning Criteria analysis and affected system analysis, voltage network upgrades identified in the MWEX voltage stability analysis, stability network upgrades identified in the MISO transient stability analysis, and short circuit network upgrades identified in the MISO short circuit analysis. The total costs of ERIS network upgrades for the 2024 scenario are summarized in Table 12-1.

Table 12-1: Summary of ERIS Network Upgrades

Category of Network Upgrades	Cost (\$)
Network Upgrades Identified in MWEX Voltage Stability analysis	\$0
Thermal Network Upgrades Identified in MISO Steady-State Analysis	\$16,360,000
Reactive Power Network Upgrades for Voltage Constraints	\$3,000,000
Network Upgrades Identified in Stability Analysis	\$0
Network Upgrades Identified in Short Circuit Analysis	\$0
Network Upgrades Identified in OTP LPC Analysis	\$0
Network Upgrades Identified in MDU LPC Analysis	\$300,000
Network Upgrades Identified in CIPCO affected system	\$0
Network Upgrades Identified in MPC affected system	\$53,800,000
Network Upgrades Identified in PJM affected system	\$28,030,000
Network Upgrades Identified in AECI affected system	\$0
Network Upgrades Identified in SPP affected system	\$85,310,870

Category of Network Upgrades	Cost (\$)
Shared Network Upgrades	\$0
Total	\$186,800,870

ERIS network upgrades are listed below.

Table 12-2: Reactive Power Network Upgrades Required for Mitigating Low Voltages at Cedar Mountain 345 kV Buses

NUs	Cost (\$)
2×75 MVar switched capacitor at Cedar Mountain 345 kV	\$3,000,000

Table 12-3: Network Upgrades Required for MWEX Voltage Stability

NUs	Miles	Cost (\$)
No MWEX NUs		\$0

Table 12-4: Thermal Network Upgrades in MISO Steady-State Analysis

Constraint	Owner	Mitigation	Cost (\$)
J748 POI-Raun 345 kV	MEC	MEC: Structure replacements. New rating projected to be 970 MVA.	\$1,500,000
Sheyenne-Lake Park 230 kV	XEL MPC OTP	OTP: Structure replacement. \$600,000. 2017 Aug. Ph3 OTP LPC upgrade	\$0 ¹
Sheyenne-Mapleton 115 kV	XEL OTP	Sheyenne-Mapleton 115 kV Rebuild. \$17.6M. 2017 Aug. Ph3 upgrade	\$0 ¹
Rutland-Fox Lake 161 kV	ITCM SMMPA	ITCM: Line is currently sag limited at 185 F. Mitigation required at 5 spans of line. Rating will increase to 219 MVA. \$1.1M	\$1,100,000
CSLTNET-Mapleton 115 kV	OTP	Replace switch at Mapleton	\$200,000
Buffalo 345-230-41.6 kV xfmr #2	OTP	Replace Buffalo transformer #2 with larger unit.	\$3,000,000
Wahpeton-Fergus Falls 230 kV	OTP MRES	OTP: Structure replacement. \$800,000. 2017 Aug. Ph3 OTP LPC upgrade	\$0 ¹
Audubon-Lake Park 230 kV	OTP	OTP: Structure replacement. \$100,000. 2017 Aug. Ph3 OTP LPC upgrade	\$0 ¹
Ellendale-Aberdeen 115 kV	MDU NWE	Ellendale-Aberdeen Jct 115 kV Uprate.	\$60,000
J302 POI-Mandan 230 kV	MDU	Uprate Mandan-J302 POI 230 kV line (new rating: 471/471 MVA [N/E]). J302/503 upgrade	\$0
J302 POI-Mandan 230 kV (P2-3)	MDU	Add a breaker at Merricourt which will eliminate tripping of both 230 kV lines for the "P23:230:MDU::MERRICOURT::2722" contingency.	\$1,500,000

Constraint	Owner	Mitigation	Cost (\$)
Merricourt-Wishek 230 kV	MDU	Rebuild/reconductor line with 795 ACSS (new rating: 478/511 MVA [N/E]).	\$9,000,000
Wabaco-Rochester 161 kV	DPC	A third party will fund the Rochester-Wabaco 161kV rebuild. \$11M. MTEP Appendix A project	\$0 ²
Wabaco-Alma 161 kV	DPC	A third party will upgrade the first section of the Wabaco-Alma 161kV line. \$6.3M	\$0 ²

Note 1: Costs of Network Upgrades required in DPP 2017 Aug West Ph3 are currently assigned to projects in DPP 2017 Aug. Phase 3 cycle. The NUS cost may be assigned to DPP 2018 Apr. projects if projects in DPP 2017 Aug. are withdrawn.

Note 2: If the agreement with the third party falls through, DPP 2018 Apr. projects will be responsible for the upgrades to Rochester-Wabaco and Wabaco -Alma lines.

Table 12-5: Network Upgrades Required for Transient Stability

Network Upgrades	Owner	Cost (\$)	Comments
Build Alexandria - Twin Brooks 345 kV line	OTP MRES	\$242,400,000	Only required in benchmark case ^{1, 2}
Four 50 MVAR capacitors at Alexandria 345 kV	MRES	\$16,000,000	Required in both benchmark and study cases ²

Note 1: Alexandria-Twin Brooks 345 kV line is only required for the benchmark case.

Note 2: DPP 2018 April West generation projects are not responsible for these stability Network Upgrades since the Network Upgrades are required for the benchmark case.

Table 12-6: Network Upgrades in Short Circuit Analysis

Constraint	Owner	Mitigation	Cost (\$)
No additional NUs			\$0

Table 12-7: OTP Local Planning Criteria Network Upgrades

Constraint	Owner	Mitigation	Cost (\$)
No additional NUs			\$0

Table 12-8: MDU Local Planning Criteria Network Upgrades

Constraint	Owner	Mitigation	Cost (\$)
Ellendale-Aberdeen 115 kV	MDU NWE	MDU: NWPS owns the line. Minor substation upgrades at MDU's Ellendale 115kV terminal. New rating for MDU's terminal is 192/192 [SN/SE].	\$250,000
J302 POI-Mandan 230 kV	MDU	Minor substation upgrades. New rating is 608/608 [SN/SE].	\$50,000

Table 12-9: CIPCO Affected System Network Upgrades

Constraint	Owner	Mitigation	Cost (\$)
No constraints			\$0

Table 12-10: MPC Affected System Network Upgrades

Constraint	Owner	Mitigation	Cost (\$)
Wilton-Winger 230	OTP MPC	MPC: Resag to 90°C. OTP: Mitigation not required for ASA	\$500,000
Drayton 230-115 kV xfmr	MPC	Replace transformer.	\$2,100,000
Prairie-Walle 230	MPC	Resag to 95°C.	\$200,000
Winger-Walle 230	MPC	Resag to 85°C.	\$500,000
MPC Steady State Voltage	-	Additional 1x75 MVar MSC at Bison 345 (601067)	\$1,500,000
MPC Steady State Voltage	-	2x40 MVar MSC at Maple River 230 (657754)	\$2,000,000
Transient Instability	-	+200 MVar SVC at Winger 230 (657758)	\$30,000,000
Transient Instability	-	+50 MVar SVC at Frontier 230 (657750)	\$15,000,000
Transient Instability	-	2x30 MVar MSC at Frontier 230 (657750)	\$2,000,000

Table 12-11: PJM Affected System Network Upgrades

Constraint	Owner	Mitigation	Total Cost (\$)	DPP Projects
Nelson;B- Electric JCT;R 345 kV	PJM	re-conductor the line, station conductor work and upgrade 2-disconnect switches	\$36,200,000	J982, J1084, J1181

Table 12-12: AECI Affected System Network Upgrades

Constraint	Mitigation Required	Owner	Cost (\$)	Generator
Green Forest-Township 69 kV line	Rebuild 2.12-mile-long segment of Green Forest to Township 69 kV line to 336 ACSR	AECI	\$2,895,000	None

Table 12-13: SPP ERIS Constraints Network Upgrades

Monitored Facility	Mitigation
Split Rock to Sioux City 230 kV	Upgrade Terminal Equipment at Sioux Falls 230 kV
Raun to S3451 345 kV	Rebuild Raun to S3451 345 kV Circuit
Split Rock to Sioux City 230 kV	Upgrade Terminal Equipment at Split Rock 230 kV

Table 12-14: SPP NRIS Constraints Network Upgrades

Monitored Facility	Mitigation
Grand Forks to Falconer 115 kV	Upgrade Terminal Equipment at Grand Forks 115 kV

Table 12-15: Shared Network Upgrades

Network Upgrades	Project Study Cycle	Projects sharing cost	MW Contribution	Total Network Upgrade Cost (\$)	Cost Responsibility
No SNUs					\$0

12.3 Cost Allocation Methodology

12.3.1 Thermal Network Upgrade Cost Allocation

The costs of thermal Network Upgrades (NU) for a set of generation projects (one or more sub-groups or entire group with identified NU) are allocated based on the MW impact from each project on the constrained facilities in the Study Case. For thermal constraints identified in the shoulder peak scenario, the MW impact is calculated using the shoulder peak post-DPP case. The MW impact on constraints identified in the summer peak scenario is calculated using the summer peak post-DPP case. With all Group Study generation projects dispatched in the Study Case, all thermal constraints will be identified and a distribution factor from each project on each constraint will be obtained.

Thermal NU cost will be allocated based on the pro rata share of the MW impact on all constraints from each project, where MW impact = DF * Gen Output of the project in the model where the constraint occurs. If the Network Upgrade alleviates multiple constrained facilities the cost is allocated based on the sum of the highest MW contribution on all of the constrained elements for the DPP project under contingency. The methodology to determine the cost allocation of thermal NU is:

$$\text{Project A cost portion of NU} = \text{Cost of NU} \times \left(\frac{\text{Max(Proj.A MW contribution on constraint)}}{\sum_i \text{Max(Proj.i MW contribution on constraint)}} \right)$$

12.3.2 Voltage Network Upgrade Cost Allocation

Voltage NU cost allocation will be determined by the pro rata share of the voltage impact each project has on the most constrained bus under the most constraining contingency. The voltage impact of each project will be calculated by locking all voltage regulating equipment in the model and then backing out each project one at a time to identify each project's impact to the constraint. In severe instances of voltage collapse where projects cannot be backed out one at a time, they will be added one at a time to determine their impact to the constraint.

12.3.3 Transient Stability Network Upgrade Cost Allocation

Transient stability driven Network Upgrades will be cost allocated based on the pro rata share of the total MW request of all the projects causing instability. The project(s) causing instability

will be determined by backing out each project one at a time to identify each project's impact to the constraint.

12.4 Cost Allocation

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Network Resource Interconnection service as of the draft System Impact Study report date.

For each thermal constraint, the maximum MW contribution (increasing flow) from each DPP project is calculated. MW contribution from one DPP project is set as zero if the project doesn't violate DPP reliability criteria for a constrained element.

For voltage network upgrades, generators which degrade voltage at the most constrained bus under the most constraining contingency will be responsible for mitigating these constraints.

Transient stability Network Upgrades are allocated based on projects causing instability. If multiple projects are causing instability, cost allocation will be based on pro rata share of total MW of all projects causing instability.

Additional NRIS Network Upgrades are allocated to the impacting NRIS projects. ERIS Network Upgrades will be allocated to the impacting projects only based on the ERIS results.

The calculated DF results, voltage impact, and MW contribution on each constraint are in Appendix K.1 for the 2024 scenario.

Finally, the cost allocation for each NU is calculated based on the contribution of each generating facility, as detailed in Appendix K.2 for the 2024 scenario.

Assuming all generating facilities in the DPP 2018 April West Area group advance, a summary of the costs for total NUs (NUs for ERIS, NRIS, and Interconnection Facilities) allocated to each generating facility is listed in Table 12-16.

Table 12-16: Summary of Total NU Costs Allocated to Each Generation Project

Project	Max Output (MW)	Total Cost of NU per Project (\$)	\$/MW	Share %
J953	1.83	\$0	\$0	0.00%
J954	1.4	\$0	\$0	0.00%
J963	9	\$0	\$0	0.00%
J967	150	\$5,559,843	\$37,066	2.17%
J975	150	\$30,964,119	\$206,427	12.06%
J982	300	\$43,031,520	\$143,438	16.76%
J1040	250	\$56,388,632	\$225,555	21.96%
J1045	20	\$1,570,784	\$78,539	0.61%
J1050	225	\$772,915	\$3,435	0.30%

Project	Max Output (MW)	Total Cost of NU per Project (\$)	\$/MW	Share %
J1072	150	\$5,559,843	\$37,066	2.17%
J1084	150	\$13,684,462	\$91,230	5.33%
J1092	100	\$4,987,653	\$49,877	1.94%
J1110	100	\$14,144,863	\$141,449	5.51%
J1124	100	\$4,996,067	\$49,961	1.95%
J1131	100	\$825,000	\$8,250	0.32%
J1132	50	\$278,091	\$5,562	0.11%
J1135	50	\$8,475	\$169	0.00%
J1164	80	\$13,441,263	\$168,016	5.23%
J1169	50	\$10,049,536	\$200,991	3.91%
J1174	165	\$12,779,829	\$77,454	4.98%
J1175	165	\$12,923,897	\$78,327	5.03%
J1181	200	\$24,831,383	\$124,157	9.67%
Total/Average	2567.2	\$256,798,174	\$78,498	100.00%

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Model Development

A.1 DPP 2018 April Generation Projects

Table A-1: DPP 2018 April West Area Projects

MISO Project Num	State	County	Trans. Owner	Point of Interconnection	ERIS Output	NRIS Output	Fuel Type	Service Type
J953	IA	Johnson	ITCM	AMIL.IOW_AFRYT	1.83	1.83	Diesel	External NRIS
J954	IA	Johnson	ITCM	AMIL.IOW_AFRYT	1.4	1.4	Solar	External NRIS
J963	IA	Cedar	ITCM	Bennett - Graham 69 kV	9	9	Diesel	NRIS
J967	MN	Mower	Xcel	Adams 345 kV	150	0	Wind	ERIS
J975	ND	Cass	OTP	Buffalo 115 kV	150	0	Wind	ERIS
J982	IA	Dickinson, Emmet	MEC	Obrien County - Kossuth 345 kV	300	0	Wind	ERIS
J1040	ND	McIntosh	MDU	Wishek Junction 230 kV	250	250	Wind	NRIS
J1045	MN	Murray	Xcel	Fenton - Chanarambie 115 kV	20	20	Battery	NRIS
J1050	IA	Boone, Hamilton	ITCM	Doud Tap 161 kV	225	225	Wind	NRIS
J1072	MN	Mower	Xcel	Adams 345 kV	150	150	Solar	NRIS
J1084	IA	Clinton	ITCM	Rock Creek 345 kV	150	150	Solar	NRIS
J1092	WI	Saint Croix	Xcel	Three Lakes 115 kV	100	100	Solar	NRIS
J1110	MN	Mower	SMMPA	North Austin 161 kV	100	100	Solar	NRIS
J1124	MN	Olmsted	SMMPA	Byron 345 kV	100	100	Solar	NRIS
J1131	IA	Scott	MEC	Sub 56 161 kV	100	100	Solar	NRIS
J1132	IA	Union	ITCM	Creston East 69 kV	50	0	Solar	ERIS
J1135	IA	Des Moines	ITCM	Huntwoods 69 kV	50	50	Solar	NRIS
J1164	MN	Rock	ITCM	Magnolia 161 kV	80	0	Solar	ERIS
J1169	SD	McCook	Xcel	Grant 115 kV	50	0	Solar	ERIS
J1174	IA	Worth	ITCM	Bison - Colby 345 kV	165	165	Solar	NRIS
J1175	IA	Worth	ITCM	Bison - Colby 345 kV	165	0	Wind	ERIS

MISO Project Num	State	County	Trans. Owner	Point of Interconnection	ERIS Output	NRIS Output	Fuel Type	Service Type
J1181	IA	Chickasaw	ITCM	Hazleton - Mitchell county 345 kV	200	0	Wind	ERIS

Table A-2: Dynamic Modeling, Collector System and Shunt Compensation Modeling for DPP West Area Projects

MISO Project #	Turbine / Inverter	Shunt Compensation	Generator Modeling	Generator Reactive Power Capability	Collector System
J963	3 CAT Diesel 2 MW	None	One 3 MW unit One 6 MW unit	For existing generation Qmin = - 1.4333 MVAR Qmax = + 1.4333 MVAR For CAT Diesel Qmin = - 2.8667 MVAR Qmax = + 2.8667 MVAR	Not Applicable
J967	60 GE 2.5 MW -116	2x14 MVAR	One 150 MW unit	Qmin = - 49.302 MVAR Qmax = + 49.302 MVAR	R = 0.007 X = 0.0123 B = 0.04553
J975	60 GE 2.5 MW -116	1x6 MVAR	One 150 MW unit	Qmin = - 72.648 MVAR Qmax = + 72.648 MVAR	R = 0.00340 X = 0.00490 B = 0.01250
J982	150 Vestas V110 2 MW	2x25.5 MVAR	Two 150 MW units	Gen 1: 150 MW Qmin = - 49.303 MVAR Qmax = + 49.303 MVAR Gen 2: 150 MW Qmin = - 49.303 MVAR Qmax = + 49.303 MVAR	Collector 1: R = 0.02885 X = 0.02659 B = 0.12103 1x25.5 MVAR Cap Collector 2: R = 0.02800 X = 0.02534 B = 0.11000 1x25.5 MVAR Cap
J1040	100 GE 2.52 MW -127	1*17 MVAR	Two 125 MW units	Gen 1: 125 MW Qmin = - 61.025 MVAR Qmax = 61.025 MVAR Gen 2: 125 MW Qmin = - 61.025 MVAR Qmax = 61.025 MVAR	Collector 1: R = 0.00547 pu X = 0.00639 pu B = 0.02249 pu 1*17 MVAR Cap Collector 2: R = 0.0102 pu X = 0.01436 pu B = 0.03942 pu
J1045	6 4.2 MVA TMEIC Ninja 840KW (5*0.84 MVA) EES	None	One 20 MW unit	Qmin = - 15.33 MVAR Qmax = + 15.33 MVAR	Not Available
J1050	90 GE 2.5 MW -116	4*15 MVAR	Two 112.5 MW units	Gen 1: 112.5 MW Qmin = - 54.4862 MVAR Qmax = + 54.4862 MVAR Gen 2: 112.5 MW Qmin = - 54.4862 MVAR Qmax = + 54.4862 MVAR	Collector 1: R = 0.0054307 pu X = 0.005201 pu B = 0.0505 pu 2*15 MVAR Collector 2: R = 0.009021 pu X = 0.0009061 pu

MISO Project #	Turbine / Inverter	Shunt Compensation	Generator Modeling	Generator Reactive Power Capability	Collector System
					B = 0.0764 pu 2*15 MVAR
J1072	41 TMEIC 4.05 MVA	2*12 MVAR	One 150 MW unit	Qmin = -49.3 MVAR Qmax = +49.3 MVAR	R = 0.0011 pu X = 0.0009 pu B = 0.0054 pu
J1084	41 TMEIC 4.05 MVA	2*12 MVAR	One 150 MW unit	Qmin = -49.3 MVAR Qmax = +49.3 MVAR	R = 0.0011 pu X = 0.0009 pu B = 0.0054 pu
J1092	152 Schneider XC680-NA	1*12 MVAR	One 100 MW unit	Qmax = +48.4 MVAR Qmin = - 48.4 MVAR	R = 0.001858 pu X = 0.001644 pu B = 0.000363 pu
J1110	34 Power Electronics HEMK FS3150KU 3.15 MVA	None	One 100 MW unit	Qmin = -48.55 MVAR Qmax = 48.55 MVAR	R = 0.00622 pu X = 0.00846 pu B = 0.02999 pu
J1124	34 Power Electronics HEMK FS3150KU 3.15MVA, 3.276 MVA derated	None	One 100 MW unit	Qmin = -48.55 MVAR Qmax = 48.55 MVAR	R = 0.00622 pu X = 0.00846 pu B = 0.0299 pu
J1131	34 Power Electronics FS3000CU15	None	One 100 MW unit	Qmin = -59.55 MVAR Qmax = 59.55 MVAR	R = 0.00776 pu X = 0.00951 pu B = 0.02263 pu
J1132	17 Power Electronics FS3000CU15	None	One 50 MW unit	Qmin = -29.7 MVAR Qmax = 29.7 MVAR	R = 0.01593 pu X = 0.01958 pu B = 0.01154 pu
J1135	17 Power Electronics FS3000CU15	None	One 50 MW unit	Qmin = -29.7 MVAR Qmax = 29.7 MVAR	R = 0.00776 pu X = 0.00951 pu B = 0.02263 pu
J1164	30 Power Electronics FS3001CU15	None	One 80 MW unit	Qmin = - 26.29 MVAR Qmax = 26.29 MVAR	R = 0.0007 pu X = 0.00112 pu B = 0.00046 pu
J1169	18 FS3001CU15	4X4 MVAR	One 50 MW unit	Qmin = - 16.86 MVAR Qmax = 16.86 MVAR	R = 0.005008 pu X = 0.006485 pu B = 0.005161 pu
J1174	62xFS3000 Power Electronics	1x4 MVAR	One 165 MW unit	Qmin = - 79.9 MVAR Qmax = 79.9 MVAR	R = 0.00661 pu X = 0.007671 pu B = 0.0920 pu
J1175	66 GE 2.5 MW -116	1X9 MVAR	One 165 MW unit	Qmin = - 79.9 MVAR Qmax = 79.9 MVAR	R = 0.00661 pu X = 0.007677 pu B = 0.0920 pu
J1181	80 GE 2.5 MW -116	2x6 MVAR	One 200 MW unit	Qmin = - 96.8644 MVAR Qmax = +96.8644 MVAR	R = 0.00340 pu X = 0.0049 pu B = 0.0125 pu

Table A-3: DPP 2018 April Central Area Projects

MISO Project Num	State	County	Trans. Owner	Point of Interconnection	ERIS Output	NRIS Output	Fuel Type	Service Type
J955	IL	Sangamon	ATXI	Austin 345 kV	1040 sum/ 1165 win	1040 sum/ 1165 win	CC	NRIS
J956	MO	Ralls	UEC	Spencer Creek 345 kV	200	200	Solar	NRIS
J968	IN	Jasper, White	NIPS	Reynolds 345 kV	200	200	Wind	NRIS
J974	IL	Fulton, Peoria	AIC	Mapleridge 345 kV	225	225	Wind	NRIS
J976	MO	Warren	UEC	Montgomery - Enon 345 kV	300	300	Solar	NRIS
J979	IL	Christian	ATXI	Pana 345 kV	170	170	Wind	NRIS
J987	MO	Montgomery	UEC	Montgomery 161 kV	100	100	Solar	NRIS
J991	IL	Clay	AIC	Xenia 345 kV	150	150	Solar	NRIS
J992	IN	Cass	DEI	Walton 230 kV	200	200	Solar	NRIS
J993	IN	Boone	IPL	Hortonville - Whitestown 345 kV	200	200	Solar	NRIS
J994	MO	Callaway	UEC	Guthrie 161 kV	100	100	Solar	NRIS
J1022	IL	McLean	AIC	Weedman 138 kV	150	0	Wind	ERIS
J1025	MO	Knox	ATXI	Zachary - Maywood 345 kV	290	290	Wind	NRIS
J1026	MO	Audrain, Ralls	UEC	Maywood - Spencer Creek 345 kV	380	265	Wind	NRIS
J1027	IN	Pike	HE	Ratts 161 kV	150	150	Solar	NRIS
J1028	IN	Pike	HE	Ratts - Victory 161 kV	150	150	Solar	NRIS
J1034	MO	Stoddard	UEC	Stoddard - Morley 161 kV	225	225	Solar	NRIS
J1055	IL	Mason	AIC	Mason 138 kV	144	144	Wind	NRIS
J1058	IN	Lake	NIPS	Schahfer - St. John 345 kV	200	200	Solar	NRIS
J1063	IN	Clinton	DEI	New London - Frankfort 230 kV	195	195	Solar	NRIS
J1067	IN	Jasper, Pulaski	NIPS	Reynolds - Burr Oak 345 kV	240	240	Solar	NRIS
J1069	IN	Jasper, Pulaski	NIPS	Reynolds 345 kV	200	200	Wind	NRIS
J1074	IN	Gibson	SIGE	Francisco 138 kV	200	200	Solar	NRIS
J1087	MO	Scott	UEC	Miner - Kelso 161 kV	200	200	Solar	NRIS
J1094	IL	Washington	AIC	Prest 138 kV	150	150	Solar	NRIS
J1096	IL	Saline	AIC	Norris City North - Muddy 138 kV	150	150	Solar	NRIS

MISO Project Num	State	County	Trans. Owner	Point of Interconnection	ERIS Output	NRIS Output	Fuel Type	Service Type
J1102	IL	Logan	AIC	Fogarty 138 kV	70	70	Solar	NRIS
J1107	MO	Cape Girardeau	UEC	Kelso - Lutesville 345 kV	200	200	Solar	NRIS
J1115	IL	Macon	AIC	Latham - Oreana 345 kV	200	200	Wind	NRIS
J1139	IL	Champaign	AIC	Sidney 138 kV	135	135	Solar	NRIS
J1145	MO	Callaway	UEC	Overton - (McCrede) - Montgomery 345 kV	250	250	Solar	NRIS
J1152	IN	Hancock, Shelby	IPL	Gwynneville - Sunnyside 345 kV	200	200	Solar	NRIS
J1180	IL	Clark	AIC	Casey West - Sullivan 345 kV	75	75	Solar	NRIS
J1182	MO	Adair	AIC	Zachary 345 kV	250	250	Solar	NRIS
J1189	IN	Brown, Martin	DEI	Crane Solar 69 kV	4.95	4.95	Battery	NRIS

Table A-4: DPP 2018 April Michigan Area Projects

MISO Project Num	State	County	Trans. Owner	Point of Interconnection	ERIS Output	NRIS Output	Fuel Type	Service Type
J984	MI	Gratiot	METC	Nelson Rd. 345 kV	200	200	Wind	NRIS
J996	MI	Lenawee	METC	Beecher - Moraco 138 kV	80	80	Solar	NRIS
J1043	MI	Montcalm	METC	Vergennes - Nelson Rd 345 kV	374.4	374.4	Wind	NRIS
J1062	MI	Washtenaw	ITCT	Majestic - Lemoyne 345 kV	150	150	Solar	NRIS
J1071	MI	Oceana	METC	Donaldson 138 kV	100	100	Solar	NRIS
J1172	MI	Genesee	METC	Dort - Garfield 138 kV	50	50	Solar	NRIS
J1178	MI	Eaton	METC	Oneida 138 kV	65	65	Solar	NRIS

Table A-5: DPP 2018 April ATC Area Projects

MISO Project Num	State	County	Trans. Owner	Point of Interconnection	ERIS Output	NRIS Output	Fuel Type	Service Type
J986	WI	Wood	ATC	Port Edwards - Sand Lake 138kV	149.76	149.76	Solar	NRIS
J1000	WI	Grant	ATC	Nelson Dewey 138 kV	50	50	Solar	NRIS
J1002	WI	Waushara	ATC	Wautoma 138 kV	99	99	Solar	NRIS
J1003	WI	Dodge	ATC	North Beaver Dam 69 kV	50	50	Solar	NRIS
J1042	WI	Walworth	ATC	North Lake Geneva 138 kV	180	180	Solar	NRIS
J1101	WI	Manitowoc	ATC	Kewaunee 138 kV	20	20	Battery	NRIS
J1153	WI	Sheboygan	ATC	Holland 138 kV	150	150	Solar	NRIS
J1154	WI	Jefferson	ATC	Jefferson 138 kV	75	75	Solar	NRIS
J1171	WI	Dodge	ATC	Butternut 138 kV	100	100	Solar	NRIS
J1183	MI	Delta	ATC	Heritage Garden	1.35	0	Solar	ERIS
J1188	WI	Rock	ATC	Sheepskin 69 kV	50	50	Solar	NRIS

A.2 Model Review Comments

Table A-6: Model Review Comments

Company	Python/ Idev File Name	2024 SH Study	2024 SH Benchmark	2024 SPK Study	2024 SPK Benchmark
ICs	J951 Comments.py	X	X	X	X
ICs	J1038 Comments.py	X	X	X	X
ICs	J1050 Comments.py	X	X	X	X
ICs	J1086 Comments.py	X	X	X	X
ICs	J1086 Comments_SH.py	X	X		
ICs	J1086 Comments_PK.py			X	X
ICs	J1092 Comments.py	X	X	X	X
ICs	J1098 Comments.py	X	X	X	X
ICs	J1106 Comments.py	X	X	X	X
ICs	J1108 Comments.py	X	X	X	X
ICs	J1109 Comments_SH.py	X	X		
ICs	J1109 Comments_PK.py			X	X
ICs	J1110 Comments.idv	X	X	X	X
ICs	J1114_Update_SH.idv	X	X		
ICs	J1114_Update_PK.idv			X	X
ICs	J1122 Comments.py	X	X	X	X
ICs	J1124 Comments.py	X	X	X	X
ICs	J1128 Comments.py	X	X	X	X
ICs	J1140 Comments.py	X	X	X	X
ICs	J1143 Comments.py	X	X	X	X
ICs	J1164 Comments.py	X	X	X	X
ICs	J1169 Comments.py	X	X	X	X
ICs	J1170 Comments.py	X	X	X	X
ICs	J1187 Comments.py	X	X	X	X
ATC	ATC_SH_Study.py	X			
ATC	ATC_SH_Bench.py		X		
ATC	ATC_PK_Study.py			X	
ATC	ATC_PK_Bench.py				X
CIPCO	CIPCO IR23_V33_SH.idv	X	X		
CIPCO	CIPCO IR23_V33_PK.idv			X	X
CIPCO	CIPCO IR24-IR34_V33_SH.IDV	X	X		

Company	Python/ Idev File Name	2024 SH Study	2024 SH Benchmark	2024 SPK Study	2024 SPK Benchmark
CIPCO	CIPCO IR24-IR34_V33_PK.IDV			X	X
MDU	MDU_Updates-DPP_2018_APR_West_Phase1_Models_190924.idv	X	X	X	X
MDU	J929_SH_Study.py	X			
MDU	J929_SH_Bench.py		X		
MDU	J929_PK_Study.py			X	
MDU	J929_PK_Bench.py				X
MEC	MEC_DPP_2018_APR_West_Ph1_SH_Updates.py	X	X		
MEC	MEC_DPP_2018_APR_West_Ph1_SUM_Updates.py			X	X
MEC	Turn off retirements.py	X	X	X	X
MEC	Correct V Control.py	X	X	X	X
MEC	Correct X 631144-41814-631139.py	X	X	X	X
MEC	Gen Correction_SH_Study.idv	X			
MEC	Gen Correction_SH_Bench.idv		X		
MEC	Gen Correction_PK_Study.idv			X	
MEC	Gen Correction_PK_Bench.idv				X
MP	J1143 POI Chng.py	X	X	X	X
MPC	MPC-fixtrngs-APR18_West_DPP-SH_study_190918.sav.idv	X	X	X	X
ICs	J1032 Chng.py	X	X	X	X
ICs	J1041 Chng.py	X	X	X	X
ICs	J1045 Chng.py	X	X	X	X
ICs	J1054 Chng.py	X	X	X	X
ICs	J1057 Chng.py	X	X	X	X
ICs	J1061 Chng.py	X	X	X	X
ICs	J1174 Chng.py	X	X	X	X
ICs	J1179 Chng.py	X	X	X	X
Changes applied to Phase 2 study					
MISO	RMV_DPP-2018-West_Ph1.py	X	X	X	X
MISO	RMV_DPP-2017Aug-West_Ph1.py	X	X	X	X
MISO	RMV_DPP-2018-ATC.py	X	X	X	X
MISO	RMV_DPP-2018-Central.py	X	X	X	X
MISO	RMV_DPP-2018-MI.py	X	X	X	X
MISO	SH-MW_Dec_DPP-2018.py	X	X		

Company	Python/ Idev File Name	2024 SH Study	2024 SH Benchmark	2024 SPK Study	2024 SPK Benchmark
MISO	PK-MW_Dec_DPP-2018.py			X	X
MISO	RMV_DPP-Prior.py	X	X	X	X
MISO	J963_Update.py	X	X	X	X
MISO	RMV_J528 NUs.py	X	X	X	X
MISO	RMV_J598 NUs.py	X	X	X	X
MISO	RMV DPP-2017Aug-Ph1_NUs.py	X	X	X	X
MISO	RMV DPP-2017Aug-Ph1_BaseCase NU.py	X	X	X	X
MISO	RMV DPP-2018Apr_BaseCase NU.py			X	X
MISO	RMV PJM Withdrawn Prjs.py	X	X	X	X
MISO	RMV SPP Withdrawn Prjs.py	X	X	X	X
MISO	Add Cap J1092.py	X	X	X	X
CIPCO	RMV CIPCO IR27.py	X	X	X	X
SPTI	Correct Areas.py	X	X	X	X
SPTI	Correct impedance.py	X	X	X	X
SPP	GEN-2016-096_POI.idv	X	X	X	X
SPP	GEN-2016-115_POI.idv	X	X	X	X
SPP	GEN-2014-021_Duplicate.idv	X	X	X	X
SPP	GEN-2015-005_Duplicate.idv	X	X	X	X
SPP	GEN-2015-007_Duplicate.idv	X	X	X	X
SPP	GEN-2016-151_Duplicate.idv	X	X	X	X
MISO	SH-MW_Dec_DPP-2018_West.py	X	X		
MISO	PK-MW_Dec_DPP-2018_West.py			X	X
MISO	RMV ATC J807_J819_J821.py	X	X	X	X
MISO	Remove NU J807 J819 J821.idv	X	X	X	X
MPC	MPC-fixrtngs-APR18_West_DPP_Ph2-ALL.idv	X	X	X	X
MDU	MDU-Updates_APR18_West_DPP_Ph2_AllModels.idv	X	X	X	X
MDU	MDU Move J580 POI.py	X	X	X	X
MDU	MDU-CON-Additions.con				
MDU	MDU_FaultFiles_DPP-2018Apr-Ph2.zip				
ATC	J732_Updates.con				
ATC	J732_Updates.swi				
ITCM	Walters Removal.idv	X	X	X	X
MEC	MEC-DPP2018-APR-West-Ph2-Updates.py	X	X	X	X

Company	Python/ Idev File Name	2024 SH Study	2024 SH Benchmark	2024 SPK Study	2024 SPK Benchmark
MEC	2281_w_mec_P42				
MEC	2286_w_mec_P42				
MEC	2288_w_mec_P42				
MEC	2289_w_mec_P42				
MEC	2298_w_mec_P55				
ATC	J1000_Change_POI.idv	X	X	X	X
J1128	J1128 Update.py	X	X	X	X
MISO	RMV recent wd prjs.py	X	X	X	X
MISO	Update J1135.py	X	X	X	X
Changes applied to Phase 3 study					
MISO	Corrections_DPP17.py	X	X	X	X
MDU	DPP_AUG17_West_Ph3_AllPreliminaryModels-MDU_Updates.idv	X	X	X	X
MISO	J897 Q Correction.py	X	X	X	X
MISO	J926 Solar_SH.py	X	X		
MISO	J926 Solar_PK.py			X	X
CIPCO	Liberty - Dundee Rating Correction.idv	X	X	X	X
MISO	Move J628-J897 POI.py	X	X	X	X
MPC	MPC-fix-praicaps.idv	X	X	X	X
MPC	MPC-fixrtngs-DPP_AUG17_West.idv	X	X	X	X
XEL	Remove Fieldon Series Cap NSP.idv	X	X	X	X
MDU	RemoveMerricourtLoad.idv	X	X	X	X
MDU	Heskett230SubRetirement.idv	X	X	X	X
MDU	Remove_Beulah-Mandan115_Rebuild-ALL.idv	X	X	X	X
MDU	Update_MerricourtWind-SH.idv	X	X		
MDU	Update_MerricourtWind-SUM.idv			X	X
OTP	Audubon-Erie ratings v33.idv	X	X	X	X
MISO	RMV Withdrawn_DPP18-West.py	X	X	X	X
MISO	RMV Withdrawn_DPP18-Central-MI.py	X	X	X	X
MISO	Update MW Reduction_SH.py	X	X		
MISO	Update MW Reduction_PK.py			X	X
MISO	RMV Withdrawn_DPP2017.py	X	X	X	X
SPTI	RMV BaseCase NUs.py	X	X	X	X

Company	Python/ Idev File Name	2024 SH Study	2024 SH Benchmark	2024 SPK Study	2024 SPK Benchmark
MISO	Adams 345-161 transformer upgrade.idv	X	X	X	X
MISO	Crookston-Falconer uprate.idv	X	X	X	X
MISO	J628897 POI-Prairie.idv	X	X	X	X
MISO	Helena-Chub Lake 2nd Circuit.py	X	X	X	X
MISO	mapleton-Sheyenne 115 kV rebuild.idv	X	X	X	X
MISO	RICE_XFMR-ratingserror.py	X	X	X	X
MISO	Rice-BeaverCreek-Adams-Harmony-CORRECT.idv	X	X	X	X
MISO	WillmarTR_V33.idv	X	X	X	X
MISO	Aug17DPPIII_Mitigations_OTP_12-3-19.idv	X	X	X	X
SPTI	RMV Withdrawn_PJM.py	X	X	X	X
SPTI	RMV Withdrawn_SPP.py	X	X	X	X
SPTI	RMV SPP NUs.py	X	X	X	X
SPTI	DPC Correction.py	X	X	X	X
MISO	Correct Split Rock-White ratings.py	X	X	X	X
CIPCO	DPP-2018-APR-West-P3-CIPCO_Line_Removal.idv	X	X	X	X
MEC	MEC_DPP_2018_West_Ph3_Updates.py	X	X	X	X
OTP	2018-Apr-Ph3 OTP model updates_v33.idv	X	X	X	X
MISO	Add 150 Mvar cap at CDRMTH.py	X	X		
MISO	J628897 POI-Prairie.py	X	X	X	X
MISO	Remove DPC NUs.py	X	X	X	X
MISO	Remove MDU NUs.idv	X	X	X	X
OTP	MR xfmr1 replacement impedance_V33.py	X	X	X	X

A.3 MISO North as the Study Sink

Table A-7: MISO North as the Study Sink

Area #	Area Name	Area #	Area Name
207	HE	608	MP
208	DEI	613	SMMPA
210	SIGE	615	GRE
216	IPL	620	OTP
217	NIPS	627	ALTW
218	METC	633	MPW
219	ITC	635	MEC
295	WEC	661	MDU
296	MIUP	663	BEPC-MISO
314	BREC	680	DPC
333	CWLD	694	ALTE
356	AMMO	696	WPS
357	AMIL	697	MGE
360	CWLP	698	UPPC
361	SIPC	701	MISO Prior
600	Xcel		

A.4 PJM Market as PJM Projects Sink

Table A-8: PJM Market as PJM Projects Sink

Area #	Area Name	Area #	Area Name
201	AP	230	PECO
202	ATSI	231	PSE&G
205	AEP	232	BGE
209	DAY	233	PEPCO
212	DEO&K	234	AE
215	DLCO	235	DP&L
222	CE	236	UGI
225	PJM	237	RECO
226	PENELEC	320	EKPC
227	METED	345	DVP
228	JCP&L	363	LGEE
229	PPL	703	PJM Prior

A.5 SPP Market as SPP Projects Sink

Table A-9: SPP Market as SPP Projects Sink

Area #	Area Name	Area #	Area Name
515	SWPA	541	KCPL
520	AEPW	542	KACY
523	GRDA	544	EMDE
524	OKGE	545	INDN
525	WFEC	546	SPRM
526	SPS	640	NPPD
527	OMPA	645	OPPD
531	MIDW	650	LES
534	SUNC	652	WAPA
536	WERE	659	BEPC-SPP
540	GMO	702	SPP Prior

A.6 Contingency Files used in Steady-State Analysis

Table A-10: List of Contingencies used in Steady-State Analysis

Contingency File Name	Description	Shoulder	Peak
Automatic single element contingencies	Single element outages at buses 60 kV and above in the study region	x	x
CC Bipole Events.con	Specified category P1, P7 contingencies in GRE Coal Creek	x	x
HVDC_Red_2024SH.con	Contingencies with HVDC reduction	x	
HVDC_Red_2024PK.con	Contingencies with HVDC reduction		x
MEC-DPP2018APRWESTPH1-Cat P1-2019.10.08.con	Specified category P1 contingencies in MEC	x	x
MEC-DPP2018APRWESTPH1-Cat P2-2019.10.08.con	Specified category P2 contingencies in MEC	x	x
MEC-DPP2018APRWESTPH1-Cat P5-2019.10.08.con	Specified category P5 contingencies in MEC	x	x
MEC-DPP2018APRWESTPH1-Cat P7-2019.10.08.con	Specified category P7 contingencies in MEC	x	x
MISO19_2024_SUM_TA_P1_ATC.con	Specified category P1 contingencies in ATC	x	x
MISO19_2024_SUM_TA_P1_IOWA.con	Specified category P1 contingencies in Iowa	x	x
MISO19_2024_SUM_TA_P1_MINN-DAKS.con	Specified category P1 contingencies in Minnesota, Dakotas	x	x
MISO19_2024_SUM_TA_P1_P2_P4_P5_NoLoadLoss.con	Specified category P1, P2, P4, P5 contingencies in MISO North	x	x
MISO19_2024_SUM_TA_P2_P4_P5_P7_LoadLoss.con	Specified category P2, P4, P5, P7 contingencies in MISO North	x	x
MISO19_2024_SUM_TA_P1_AMEREN.con	Specified category P1 contingencies in Ameren	x	x
2020_RTEP_Single_2017Aug-Updated-3-4-19.con	Specified single contingencies in PJM	x	x
2020_RTEP_Bus_AC2-updated-3-4-19.con	Specified bus contingencies in PJM	x	x
2020_RTEP_Line_FB_2017Aug-updated-3-4-19.con	Specified breaker failure contingencies in PJM	x	x
2020_RTEP_Tower_AC2-updated-3-4-19.con	Specified common structure contingencies in PJM	x	x
160303-KACY_P1.con	Specified category P1 contingencies in KACY	x	x
160303-KACY_P2.con	Specified category P2 contingencies in KACY	x	x
AECI-AMMO.CON	Specified contingencies between AECI and AMMO	x	x
KCPL_P1.con	Specified category P1 contingencies in KCPL	x	x
KCPL_P2.con	Specified category P2 contingencies in KCPL	x	x
KCPL_P4.con	Specified category P4 contingencies in KCPL	x	x

Contingency File Name	Description	Shoulder	Peak
KCPL_P5.con	Specified category P5 contingencies in KCPL	x	x
KCPL_P7.con	Specified category P7 contingencies in KCPL	x	x

Model Data

B.1 Power Flow Model Data

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B.2 Dynamic Model Data

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B.3 2024 Slider Diagrams

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Appendix

C

Reactive Power Requirement Analysis Results (FERC Order 827)

Table C-1: Reactive Power Requirement Analysis Results

Project #	Type	HV Side Bus #	Lagging Power Factor Results				Leading Power Factor Results				Inverter Inherent Power Factor	Shunt Compensation
			MW from plant to HV side (P)	MVAR from plant to HV side (Q)	Lagging Power Factor at HV Side	Meet Lagging Power Factor Req.?	MW from plant to HV side (P)	MVAR from plant to HV side (Q)	Leading Power Factor at HV Side	Meet Leading Power Factor Req.?		
J967	Wind	89673	146.82	49.43	0.9477	Yes	146.60	-77.38	-0.884	Yes	± 0.95	2×14 MVAR cap bank on 34.5 kV system
J975	Wind	89753	147.53	49.23	0.9486	Yes	147.05	-108.29	-0.805	Yes	± 0.90	1×6 MVAR cap bank on 34.5 kV system
J982	Wind	89823	283.54	88.27	0.9548	No	281.87	-161.69	-0.867	Yes	± 0.95	1×25.5 MVAR cap bank on each of the two 34.5 kV system
J1040	Wind	40403	244.51	83.83	0.9459	Yes	243.53	-186.97	-0.793	Yes	± 0.90	1×17 MVAR cap bank on 34.5 kV system
J1045	Battery	88742	19.85	13.49	0.8271	Yes	19.83	-17.39	-0.752	Yes	± 0.794	None
J1050	Wind	40503	219.98	81.70	0.9374	Yes	219.41	-158.28	-0.811	Yes	± 0.90	2×15 MVAR cap bank on each of the two 34.5 kV system
J1072	Solar	40723	148.12	46.10	0.9548	No	147.95	-78.84	-0.883	Yes	± 0.95	2×12 MVAR cap bank on 34.5 kV system
J1084	Solar	40842	148.14	46.56	0.9540	No	147.98	-78.13	-0.884	Yes	± 0.95	2×12 MVAR cap bank on 34.5 kV system
J1092	Solar	40923	99.03	38.84	0.9310	Yes	98.87	-74.33	-0.799	Yes	± 0.90	1 x 12 MVAR cap bank on 34.5 kV system
J1110	Solar	41103	98.53	32.35	0.9501	No	98.26	-68.47	-0.820	Yes	± 0.90	None
J1124	Solar	41243	98.54	32.04	0.9510	No	98.25	-69.04	-0.818	Yes	± 0.90	None
J1131	Solar	41313	97.99	37.36	0.9344	Yes	97.4500	-88.4200	-0.7406	Yes	± 0.86	None
J1132	Solar	41323	48.96	19.79	0.9271	Yes	48.77	-41.58	-0.761	Yes	± 0.86	None
J1135	Solar	41353	49.20	21.03	0.9195	Yes	49.09	-39.89	-0.776	Yes	± 0.86	None
J1164	Solar	41643	79.31	13.55	0.9857	No	79.24	-40.34	-0.891	Yes	± 0.95	None
J1169	Solar	41693	49.38	24.61	0.8950	Yes	49.41	-24.23	-0.898	Yes	± 0.95	4×4 MVAR cap bank on 34.5 kV system
J1174	Solar	41743	161.51	57.83	0.9415	Yes	161.18	-108.67	-0.829	Yes	± 0.90	1×4 MVAR cap bank on 34.5 kV system
J1175	Wind	41753	161.13	61.97	0.9334	Yes	160.63	-110.45	-0.824	Yes	± 0.90	1×9 MVAR cap bank on 34.5 kV system
J1181	Wind	41813	196.28	67.70	0.9453	Yes	195.64	-144.36	-0.805	Yes	± 0.90	2×6 MVAR cap bank on 34.5 kV system

2024 Summer Peak Contingency Analysis Results

D.1 Constraints in 2024 Summer Peak (SPK) Condition

Table D-1: 2024 SPK System Intact Thermal Constraints

Table D-2: 2024 SPK System Intact Voltage Constraints

Table D-3: 2024 SPK Category P1 Thermal Constraints

Table D-4: 2024 SPK Category P1 Voltage Constraints

Table D-5: 2024 SPK Category P2-P7 Thermal Constraints

Table D-6: 2024 SPK Category P2-P7 Voltage Constraints

Table D-7: 2024 SPK Non-Converged Contingencies

Table D-8: 2024 SPK Non-Converged Contingencies DCCC Results

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2024 Summer Shoulder Contingency Analysis Results

E.1 2×75 MVar Switched Capacitor Justification Results

Table E-1: 2×75 MVar Switched Capacitor at Cedar Mountain Justification
Results

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E.2 Stage-2 Contingency Analysis with Base Case NUs

Table E-2: Stage-2 SH System Intact Thermal Constraints

Table E-3: Stage-2 SH System Intact Voltage Constraints

Table E-4: Stage-2 SH Category P1 Thermal Constraints

Table E-5: Stage-2 SH Category P1 Voltage Constraints

Table E-6: Stage-2 SH Category P2-P7 Thermal Constraints

Table E-7: Stage-2 SH Category P2-P7 Voltage Constraints

Table E-8: Stage-2 SH Non-Converged Contingencies

Table E-9: Stage-2 SH Non-Converged Contingencies DCCC Results

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Local Planning Criteria Analysis Results

F.1 OTP LPC Analysis

Below is the OTP local planning criteria analysis report.

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F.2 MDU LPC Analysis

Below is the MDU local planning criteria analysis report.

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Affected System Contingency Analysis Results

G.1 CIPCO Affected System Analysis Results

Table G-1: 2024 SPK CIPCO Affected System Analysis Results

Table G-2: 2024 SH CIPCO Affected System Analysis Results

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G.2 MPC Affected System Analysis Results

Below is the Affected System Analysis report provided by MPC.

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G.3 PJM Affected System Study Results

Below is the PJM affected system study report provided by PJM.

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G.4 AECI Affected System Study Results

Below is the AECI affected system study report provided by AECI.

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G.5 SPP Affected System Study Results

Below is the SPP affected system study report provided by SPP.

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Transient Stability Results

H.1 2024 Summer Peak Stability Results

Stability simulation was performed in the 2024 summer peak (PK) Phase 3 stability model without any steady state ERIIS Network Upgrades identified in DPP 2018 April Phase 3.

H.1.1 2024 SPK Stability Summary

Summer peak stability study results are summarized in Table H-1.

Table H-1: 2024 Summer Peak Phase 3 Stability Analysis Results Summary

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H.1.2 2024 SPK Stability Plots

Plots of stability simulations for 2024 summer peak Phase 3 study case are in separate files which are listed below:

AppendixH1_2024SPK_DPP 2018Apr-West_Ph3_Study_Plots.zip

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H.2 2024 Summer Shoulder Stability Results

Stability simulation was performed in the 2024 summer shoulder stability case described in Section 7.2.2.

H.2.1 2024 SH Stability Summary

Stability study results are summarized in Table H-2.

Table H-2: 2024 Summer Shoulder Phase 3 Stability Analysis Results Summary

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H.2.2 2024 SH Stability Plots

Plots of stability simulations for 2024 summer shoulder Phase 3 study case are in separate files which are listed below:

AppendixH2_2024SH_DPP 2017Feb-West_Ph3_Study_Plots.zip

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MWEX Voltage Study Details

Below is the MWEX voltage stability study report provided by ATC.

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Short Circuit Analysis

J.1 J967 & J1072 Short Circuit Study

J.2 J975 Short Circuit Study

J.3 J982 Short Circuit Study

J.4 J1040 Short Circuit Study

J.5 J1045 Short Circuit Study

J.6 J1050 Short Circuit Study

J.7 J1084 Short Circuit Study

J.8 J1092 Short Circuit Study

J.9 J1110 Short Circuit Study

J.10 J1124 Short Circuit Study

J.11 J1131 Short Circuit Study

J.12 J1132 Short Circuit Study

J.13 J1135 Short Circuit Study

J.14 J1164 Short Circuit Study

J.15 J1169 Short Circuit Study

J.16 J1174 & J1175 Short Circuit Study

J.17 J1181 Short Circuit Study

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2024 Cost Allocation Results

K.1 Distribution Factor (DF), Voltage Impact, and MW Contribution Results for Cost Allocation in 2024

Table K-1: Distribution Factor and MW Contribution on Constraints for Thermal
NU Cost Allocation

Table K-2: Voltage Impact on MISO Voltage NUs Cost Allocation

Table K-3: Voltage Impact on MISO Stability Voltage NUs Cost Allocation

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K.2 Cost Allocation Details

Table K-4: Network Upgrades Cost Allocation in 2024

Table K-4: Network Upgrades Cost Allocation in 2024																																				
Monitored Element						English Name	Owner	Cost	J953	J954	J963	J967	J975	J982	J1040	J1045	J1050	J1072	J1084	J1092	J1110	J1124	J1131	J1132	J1135	J1164	J1169	J1174	J1175	J1181	Upgrade for					
87487 J748PO1	345	635200	RAUN	3	345	1	J748 POI-Raun 345 kV	MEC	\$1,500,000	\$0	\$0	\$0	\$0	\$0	\$1,500,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	MISO SH				
Lakefield-Huntley 345 kV						Lakefield-Huntley 345 kV	ITCM	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	MISO SH			
602006 SHEYNNE4	230	620337	LAKE PARK T4	230	1	Sheyenne-Lake Park 230 kV	XEL MPC OTF	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	MISO SH				
603018 SHEYNNE7	115	620203	MAPLTN 7	115	1	Sheyenne-Mapleton 115 kV	XEL OTF	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	MISO SH				
613370 RUTLAND5	161	631042	FOX LK 5	161	1	Rutland-Fox Lake 161 kV	ITCM SMMPA	\$1,100,000	\$0	\$0	\$0	\$0	\$0	\$1,100,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	MISO SH			
620180 CSLTNET7	115	620203	MAPLTN 7	115	1	CSLTNET-Mapleton 115 kV	OTF	\$200,000	\$0	\$0	\$0	\$0	\$0	\$200,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	MISO SH			
620258 BUFFALO7	115	620358	BUFFALO3	345	620858	Buffalo 345-230-41.6 kV xfmr #2	OTF	\$3,000,000	\$0	\$0	\$0	\$0	\$3,000,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	MISO SH			
BUFFALO 9X 41.6 2																																				
620329 WAHPETN4	230	658109	FERGSFL4	230	1	Wahpeton-Fergus Falls 230 kV	OTF NRIS	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	MISO SH			
620336 AUDUBON4	230	620337	LAKE PARK T4	230	1	Audubon-Lake Park 230 kV	OTF	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	MISO SH			
661027 ELLENDL7	115	660000	ABERDN JCT 7	115	1	Ellendale-Aberdeen 115 kV	MDU NWE	\$60,000	\$0	\$0	\$0	\$0	\$0	\$0	\$60,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	MISO SH, MDU LPC			
661053 MANDAN 4	230	83021	J302 POI	230	1	J302 POI-Mandan 230 kV	MDU	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	MISO SH, MDU LPC			
Add a breaker at Merricourt which will eliminate tripping of both 230 kV lines for the "P23:230:MDU::MERRICOURT::2722" contingency						Add a breaker at Merricourt which will eliminate tripping of both 230 kV lines for the "P23:230:MDU::MERRICOURT::2722" contingency	MDU	\$1,500,000	\$0	\$0	\$0	\$0	\$0	\$1,500,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	MISO SH, MDU LPC		
661093 MERRCRT4	230	661094	WISHEK 4	230	1	Merricourt-Wishek 230 kV	MDU	\$9,000,000	\$0	\$0	\$0	\$0	\$0	\$0	\$9,000,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	MISO SH		
681532 WABACO 5	161	681537	ROCHSTR5	161	1	Wabaco-Rochester 161 kV	DPC	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	MISO SH		
681532 WABACO 5	161	681543	ALMA 5	161	1	Wabaco-Alma 161 kV	DPC	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	MISO SH		
2x75 Mvar switched capacitor at Cedar Mountain 345 kV (615647)						2x75 Mvar switched capacitor at Cedar Mountain 345 kV (615647)	GRE	\$3,000,000	\$0	\$0	\$0	\$0	\$186,441	\$584,746	\$991,525	\$144,068	\$364,407	\$0	\$0	\$0	\$0	\$0	\$0	\$50,847	\$8,475	\$211,864	\$144,068	\$84,746	\$228,814	\$0	\$0	\$0	MISO Voltage			
Stability voltage NU						Benchmark voltage NU	Multiple	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	MISO Stability		
661027 ELLENDL7	115	660000	ABERDN JCT 7	115	1	Ellendale-Aberdeen 115 kV (MDU LPC)	MDU NWE	\$250,000	\$0	\$0	\$0	\$0	\$0	\$250,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	MDU LPC		
661053 MANDAN 4	230	83021	J302 POI	230	1	J302 POI-Mandan 230 kV (MDU LPC)	MDU	\$50,000	\$0	\$0	\$0	\$0	\$0	\$0	\$50,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	MDU LPC		
Upgrade Pleasant Valley 345/161 kV Transformer #1						Upgrade Pleasant Valley 345/161 kV Transformer #1	GRE	\$3,000,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$3,000,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	NRIS	
Upgrade Pleasant Valley 345/161 kV Transformer #2						Upgrade Pleasant Valley 345/161 kV Transformer #2	GRE	\$3,000,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$3,000,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	NRIS
Green Forest-Township 69 kV line						Green Forest-Township 69 kV line	AECI	\$2,895,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	AECI	
Wilton-Winger 230 kV line						Wilton-Winger 230 kV line	MPC	\$500,000	\$0	\$0	\$0	\$0	\$206,016	\$0	\$293,984	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	MPC	
Drayton 230-115 kV xfmr						Drayton 230-115 kV xfmr	MPC	\$2,100,000	\$0	\$0	\$0	\$0	\$0	\$0	\$2,100,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	MPC	
Prairie-Walle 230 kV line						Prairie-Walle 230 kV line	MPC	\$200,000	\$0	\$0	\$0	\$0	\$0	\$0	\$200,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	MPC	
Winger-Walle 230 kV line						Winger-Walle 230 kV line	MPC	\$500,000	\$0	\$0	\$0	\$0	\$0	\$0	\$500,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	MPC	
1x75 MVar MSC at Bison 345						1x75 MVar MSC at Bison 345	MPC	\$1,500,000	\$0	\$0	\$0	\$0	\$390,582	\$0	\$1,109,418	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	MPC	
2x40 MVar MSC at Maple River 230						2x40 MVar MSC at Maple River 230	MPC	\$2,000,000	\$0	\$0	\$0	\$0	\$958,199	\$0	\$1,041,801	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	MPC	
+200 MVar SVC at Winger 230						+200 MVar SVC at Winger 230	MPC	\$30,000,000	\$0	\$0	\$0	\$0	\$12,259,887	\$0	\$17,740,113	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	MPC	
+50 MVar SVC at Frontier 230						+50 MVar SVC at Frontier 230	MPC	\$15,000,000	\$0	\$0	\$0	\$0	\$6,129,944	\$0	\$8,870,056	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	MPC	
2x30 MVar MSC at Frontier 230						2x30 MVar MSC at Frontier 230	MPC	\$2,000,000	\$0	\$0	\$0	\$0	\$817,326	\$0	\$1,182,674	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	MPC	
Nelson/B- Electric JCT/R 345 kV						Nelson/B- Electric JCT/R 345 kV	PJM	\$36,200,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$7,510,000 PJM	
SPP ERIS Network Upgrades						SPP ERIS Network Upgrades	SPP	\$85,927,678	\$0	\$0	\$0	\$4,631,343	\$6,336,833	\$17,716,774	\$11,012,024	\$1,426,716	\$0	\$4,631,343	\$0	\$3,476,653	\$3,383,313	\$3,401,667	\$0	\$0	\$0	\$0	\$6,320,652	\$4,420,468	\$6,897,617	\$6,897,617	\$4,668,050	\$0	\$0	SPP		
SPP NRIS Network Upgrades						SPP NRIS Network Upgrades	SPP	\$90,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	SPP
Total Cost Per Project for Actual NRIS Elections for each Project						Total Cost Per Project for Actual NRIS Elections for each Project		\$192,800,870	\$0	\$0	\$0	\$4,631,343	\$30,485,028	\$29,031,520	\$55,991,595	\$1,570,784	\$364,407	\$4,631,343	\$12,390,000	\$3,476,653	\$9,383,313	\$3,401,667	\$0	\$50,847	\$8,475	\$6,532,516	\$4,564,536	\$6,982,363	\$7,126,431	\$12,178,050						

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